I-Card: A Generative AI-Supported Intelligent Design Method Card Deck

Liuqing Chen
College of Computer Science and Technology
Zhejiang University
Hangzhou, Zhejiang, China
chenlq@zju.edu.cn

Wengteng Cheang
College of Computer Science and Technology
Zhejiang University
Hangzhou, Zhejiang, China
cheangwt@zju.edu.cn

Zhaojun Jiang
College of Computer Science and Technology
Zhejiang University
Hangzhou, Zhejiang, China
22321008@zju.edu.cn

Yuan Xu
School of Computer Science and Engineering
Central South University
Changsha, Hunan, China
xuan@csu.edu.cn

Zebin Cai
College of Computer Science and Technology
Zhejiang University
Hangzhou, Zhejiang, China
caizebin@zju.edu.cn

Lingyun Sun
International Design Institute
Zhejiang University
Hangzhou, Zhejiang, China
sunly@zju.edu.cn

Peter Childs
Dyson School of Design Engineering
Imperial College London
London, United Kingdom
p.childs@imperial.ac.uk

Ji Han
Department of Innovation, Technology and Entrepreneurship
University of Exeter
Exeter, United Kingdom
j.han2@exeter.ac.uk

Preben Hansen
Dept of Computer and Systems Sciences
Stockholm University
Kista, Sweden
preben@dsv.su.se

Haoyu Zuo*
Dyson School of Design Engineering
Imperial College London
London, United Kingdom
hz2019@ic.ac.uk

Abstract

A design method card deck helps designers understand and provoke thinking by presenting each method in a simple format and allow designers to switch between methods seamlessly by maintaining the same simple format across the deck. However, recent observations have shown designers hesitate to use a card deck due to the lack of support, while other tools have provided identified support with generative AI. Through a formative study, we identified the specific support designers need when applying the design method cards and intentions in integrating generative AI. Accordingly, we developed the intelligent design method card deck, I-Card, which integrates generative AI to provide applicable design methods, design knowledge and data support, and interactive and dynamic support. A user study demonstrates that I-Card improved the design efficiency and applicability by offering personalized guidance, enhanced decision-making with comprehensive data generation and provided more design inspiration via interactive support.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; **Systems and tools for interaction design**.

Keywords

Design method, design method cards, design cards, generative AI, design support tool

ACM Reference Format:
Liuqing Chen, Wengteng Cheang, Zhaojun Jiang, Yuan Xu, Zebin Cai, Lingyun Sun, Peter Childs, Ji Han, Preben Hansen, and Haoyu Zuo. 2025. I-Card: A Generative AI-Supported Intelligent Design Method Card Deck. In *CHI Conference on Human Factors in Computing Systems (CHI '25), April 26–May 01, 2025, Yokohama, Japan.* ACM, New York, NY, USA, 22 pages. https://doi.org/10.1145/3706598.3713934

*Corresponding author

This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International License.
CHI '25, Yokohama, Japan
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1394-1/25/04
https://doi.org/10.1145/3706598.3713934

1 Introduction

Design methods are systematic approaches used to guide the design process, offering frameworks and strategies that help designers structure their thinking and problem-solving efforts. These methods are essential for transferring design method knowledge into practical, effective, and innovative solutions. To implement these

design methods, tools such as design toolkits [52], interactive online platforms, digital design tools [55], method guides and workbooks and design method cards [62] have been developed [32].

Design method cards are typically organized into a card deck, such as the SUTD-MIT Design Methods Cards [7] and IDEO Design Cards [28], which includes multiple methods in a deck. Within the deck, each method card presents a method in a simple format, helping designers quickly understand and provoke thinking. Across the deck, the consistent structure reduces the cognitive load by avoiding adjusting to a different structure for a different method [14, 46, 56], this is beneficial since different methods are often applied to address different challenges in the design process. Unlike other design tools that integrate only a few methods [20], requiring designers to switch between different tools, the consistent structure makes it easier for designers to switch between methods within one tool - the card deck. Recent efforts have been made on creating more specialized cards [21, 34, 44, 64], exploring new frameworks [14, 39, 51], better integrating these cards into the design activities [19, 24, 62] and improving the awareness and applicability of design method cards by systematically reviewing design method cards [27, 49, 62].

Despite its advantages, recent observations reveal that designers still hesitate to use design method cards, as designers reported a lack of support while using these cards [27]. Other design method based tools have identified designers' support need such as understanding [31, 63], transferring [8–10, 53, 54], customizing [22, 35] and visualizing [15, 18, 37] while applying design method and provide support using generative AI. However, the specific support requirements for design method cards have not yet been fully identified, nor has generative AI been explored as a solution in this context. Therefore, this study seeks to explore the following research questions:

- **RQ1:** What specific support is required and can generative AI be introduced when applying design method cards?
- **RQ2:** How can generative AI-supported design method cards be designed?
- **RQ3:** How do designers apply these generative AI-supported design method cards in the design process?

To answer the first question, we conducted a formative study that identified the specific supports, and the potential of integrating generative AI with design method cards by examining the intentions and expected outcomes that designers associate with its use. Designers expressed the need for more actionable guidance and examples to help them apply the cards effectively in various design contexts. Additionally, they highlighted the need for more comprehensive data and detailed information to support decision-making, as the current content was too generalized. Designers expressed interests in using generative AI to provide context-specific insights and supplement the cards by answering their specific questions.

To address these difficulties and expectations, we developed the I-Card through five rounds of iteration, which eventually include five types of cards: Info Card, Method Card, Solution Card, QA (question and answer) Card, and Resource Card. These cards integrate generative AI to provide personalized design method and solution that adapts to specific project requirements, offer enhanced access to supportive knowledge and data resources, and enable dynamic interaction for improved engagement.

Next, we conducted a user study where participants engaged with the finalized I-Card and provided feedback on their experience. In this study, sixteen designers completed two design tasks using I-Card and a set of design method cards. Findings showed that, the developed characteristics of I-Card, improved the design efficiency and applicability, provided more design inspiration and supported design decision-making.

The contributions are three-fold:

- **How can generative AI contribute to supporting design method cards:** The formative study identified the specific supports required and designers' intentions and expected outcomes in integrating generative AI, including seeking more actionable guidance and example, better decision support with data, and enhanced interactivity.
- **Generative AI integrated I-Card:** The I-Card has been designed and developed, which integrates generative AI to support the applicability of design methods, offering design knowledge and data support, and improving interactive and dynamic engagement through five specific cards.
- **The effectiveness of I-Card:** The findings from user study demonstrate that integrating generative AI supports design method cards by improving design efficiency and applicability, providing more design inspiration and supported design decision-making. The findings suggest that I-Card could enhance the applicability in the design community.

2 Related Work

In this section, we first review prior work on the current landscape of tool-based design methods, identifying key gaps and opportunities. We then discuss the unique strengths of design method cards as a practical tool for presenting design methods. Finally, we explore the potential for enhancing design method cards with generative AI.

2.1 Design Methods

Design methods refer to systematic approaches that structure the design activity, guiding designers through their work by providing clear procedures and techniques [17, 23]. These methods help designers organize activities and integrate them into a cohesive design approach. Design methods can take various forms, from formal techniques used in design practices, including formalized versions of informal techniques traditionally used by designers, to adaptations from fields such as operational research, decision theory, or management sciences [27] Each method serves a specific function and is applied at different stages of the design process [62].

A variety of design methods have been developed to support diverse design activities, addressing different objectives and challenges in the design process. To better organize their application, design methods can be categorized into discover, define, develop, and deliver phases and different method should be applied into different phases [19]. In the discover phase, designers gather insights and understand the problem space. For example, user research and contextual inquiry are used to explore user needs and behaviors, with contextual inquiry observing users in their natural environment to gain insights into their workflows and pain

points [26]. In the define phase, methods like KANO models [5] and personas [47] help organize insights and identify key patterns. For instance, KANO models classify features based on customer satisfaction, while personas represent archetypal users and their needs. In the develop phase, designers generate and refine potential solutions. TRIZ (Theory of Inventive Problem Solving) [4] and concept testing [42] are examples of methods used in this phase. TRIZ identifies contradictions, a situation where improving one feature of a product leads to the degradation of another feature, and uses the contradiction matrix to identify principles that help find solutions to balance these conflicting features effectively. Concept testing evaluates potential solutions by gathering early feedback from target users, ensuring ideas are practical and meet user needs before full development. In the deliver phase, the focus is on finalizing and implementing the solution. Methods like usability testing [43] and A/B testing [33] ensure the solution performs effectively. Usability testing observes users interacting with prototypes to refine and enhance usability. A/B testing compares different design versions to identify the most effective one based on user response.

The complexity and large amount of design methods make it difficult to learn and apply, thus design methods are commonly integrated in various tools that make them accessible and usable across different design activities. Design toolkits, such as the Design Thinking Toolkit [29], offer a set of methods and resources, guiding designers through different phases of the design process with templates and worksheets. Similarly, interactive online platforms, such as Miro and Mural, provide digital templates for methods like brainstorming and journey mapping, enabling collaborative and remote application. Digital design tools, such as Figma and Canva, offer method templates for frameworks like personas and wireframing, allowing for quick implementation without building from scratch. Method guides and workbooks, such as LUMA Institute's Innovating for People [30], provide step-by-step instructions and exercises, helping designers systematically apply methods while understanding key principles. However, these tools have limitations. Many of them only offer a limited number of methods, which may not cover the variety of challenges designers face in different contexts and require designers to switch between different tools for different tasks. Since methods are often presented differently across tools, with each tool offering its own version of a method, designers may find it difficult to switch between tools or apply methods consistently across different platforms, which can increase cognitive load and interrupt the flow of work [16, 49, 59]. To address this, Design method cards, such as the IDEO Design Cards [28] and SUTD-MIT Design Methods Cards [7], break down numerous methods into concise steps and provide designers with consistent structure of instructions, making it easier to apply these methods.

2.2 Design method cards

Design method cards are typically organized into a card deck. Within the deck, each design method card serves to make the design method more accessible and applicable by organizing information into a simple, tangible and easy-to-manipulate format. Each card typically consists of key components such as the name of the design method, a brief description explaining its purpose, and step-by-step instructions for applying the method [49]. These components provide a structured and practical approach, allowing designers to quickly understand and provoke thinking.

Design method cards have been the subject of various research in the design community. These research include the development of specialized cards tailored to specific areas such as social resilience [44] inclusiveness [21], and emerging technologies like artificial intelligence [36, 60] and physical computing [41, 45, 48], addressing distinct challenges within those fields. For instance, AI-VIS EthiCards [60] help explore ethical considerations in AI and data visualization through six themes and structured activities, while AI-DEC [36] cards support user-centered design of AI explanations tailored to workplace needs. Similarly, MCVT Cards [45] and Maker Cards [41, 48] provide tools for inclusive and physical computing education, enabling teachers and students to co-design activities and empowering students to create meaningful artifacts independently. New frameworks have also been developed to better integrate design method cards into the design workflow, ensuring that they are more practical and relevant for design practices [19, 24, 39, 51]. Additionally, the research also include systematic reviews to broaden the range of design method cards available to the community, making previously unexplored options more accessible [27, 62].

However, recent observations [27] results indicate that many designers with the experience of using the card would not use them again. The remaining users tend to rely on the cards they are already familiar with, showing hesitation toward exploring new or different types. The analyses from designers reported the main barrier is the limited design method related support while using the card. Designers [2, 27, 62] have highlighted a support need for clear guidelines and actionable examples to improve the effectiveness of design method cards in various contexts. Another support requirement includes providing basic instructions for their use, detailed explanations of their roles within design processes, and features that allow for customization based on specific project needs. While Hsieh et al. [27] has begun to identify and summarize these support needs, a more detailed exploration is necessary to thoroughly understand and address these specific challenges.

In contrast, recent advancements in other design method-based tools have progressed further in identifying key support needs for designers when applying design methods. These needs include understanding the method's purpose [31, 63], transferring knowledge into actionable steps [8, 10, 53, 54], customizing methods for specific contexts [22, 35], and visualizing complex processes to enhance comprehension [11, 13, 15, 18, 37]. Generative AI has been introduced in these tools to address these needs by providing personalized content, automating customization, and generating visual representations. Generative AI provides deeper and more interactive support, helping designers bridge the gap between abstract methods and practical use, making the application of design tools smoother [12, 61]. Recent advances in identifying support needs for general design method-based tools and integrating AI into these processes have led to developments. However, these developments do not directly translate to design method cards. These cards often require more tailored, context-specific support that addresses both the physical and interactive elements unique to this format, which broader tools may not adequately provide. We aim to identify the specific support needs and challenges of using design method card

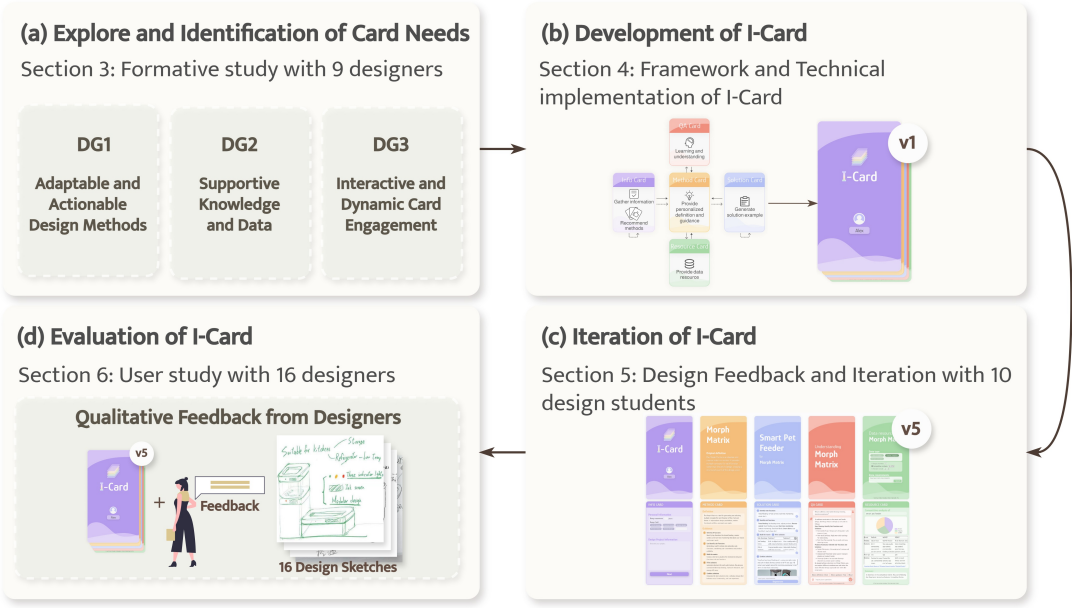


Figure 1: An overview of the study procedure for exploring, developing, iterating and evaluating the I-Card framework. The process comprises four main stages: (a) Conducting a formative study to identify designer needs and establish three design goals for AI-supported design method cards; (b) Proposing the I-Card framework and developing the first prototype by integrating generative AI; (c) Refining the I-Card prototype through five iterations based on feedback; and (d) Evaluating the I-Card in a user study to gather qualitative feedback and assess its impact on design practices.

in our formative study and then explore the potential of generative AI to address these effectively.

2.3 Study Procedure Overview

Our study consisted of four main stages, which are illustrated in Figure 1. Firstly, we conducted a formative study with 9 designers to identify support needs when using design method cards and explore how generative AI can address these issues, with the aim of deriving design goals for creating AI-supported design method cards (Figure 1a). Secondly, we proposed and developed the I-Card framework based on the design goals identified. I-Card integrated generative AI into design method cards to provide adaptive and dynamic design support tailored to specific design projects (Figure 1b). Thirdly, to refine I-Card, we iteratively gathered feedback from 10 design students and updated I-Card prototype through five rounds of iteration, ensuring alignment with user needs and enhancing usability (Figure 1c). Finally, we conducted a user study with 16 designers to evaluate designers' perceptions of finalized I-Card and assess its impact on design practice (Figure 1d).

3 Formative Study

To better understand the specific support requirements of applying design method cards and how designers intend to use generative AI to support with these cards, we conducted a formative study using design method cards and generative AI (ChatGPT) to complete a design task.

Specifically, the main goal of this study is to:

- Identify the specific challenges and required supports when using design method cards and explore the potential for integrating generative Al to provide support.
- Explore designers' intentions in employing generative AI for the application of design method cards, the expected outcomes of such integration, and the potential issues that may occur in this process.

3.1 Participants

We recruited participants through online recruitment in a design-focused community at a university. Nine participants were involved, divided into three teams of two participants (labeled as **T**1 to **T**3) each and three individuals participants(labeled as **P**1 to **P**3), to ensure that different types of design method cards could be applied across both team-based and individual contexts. Their design experience ranged from two to seven years (M = 4.67). Participants had diverse design backgrounds, including UI/UX design, VR design, product design, and service design. The participants (self-identified as seven males, two females) were aged 19 to 25 years (M = 22.44, SD = 2.454). Among the participants, three had prior experience with design method cards, three had heard of them, and two were unfamiliar. Eight participants used generative AI daily, while one used it weekly.

3.2 Procedure

We first introduced the formative study to the participants (10 minutes), then we asked participants to perform a design task with generative AI (60 minutes), and concluded with a semi-structured

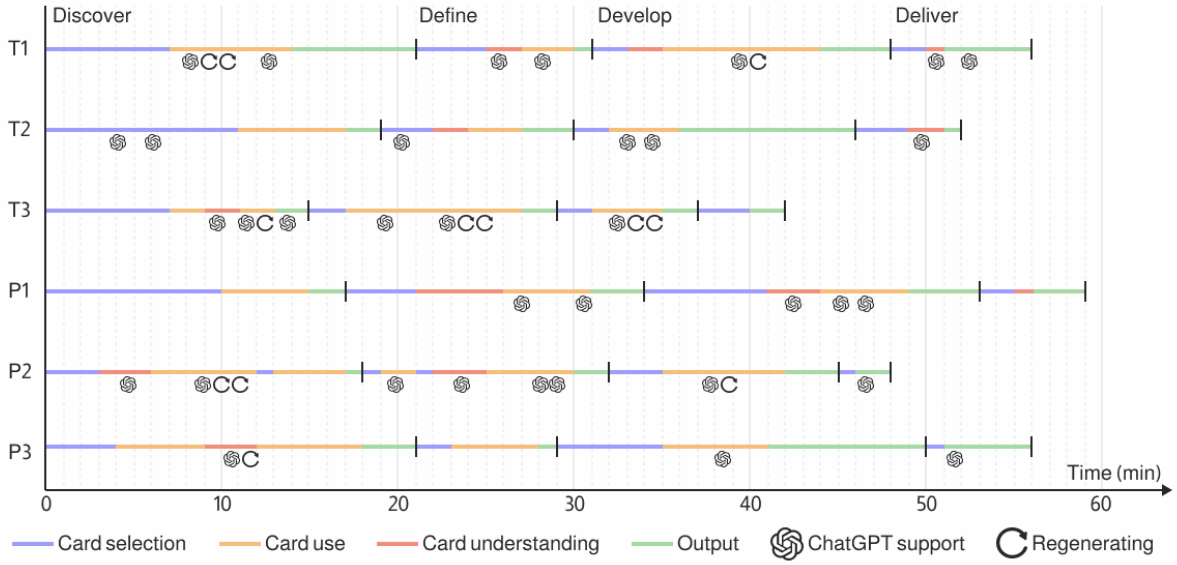


Figure 2: Formative study action log visualization.

interview about their experience (20 minutes). The study was approved by the local research ethics committee.

We selected the SUTD-MIT Design Methods Cards (referred to as SUTD-MIT Cards) [7] for the study as they provide a comprehensive and systematic representation of design methodologies. Each participant was provided with a online collaborative platform Miro and ChatGPT (GPT-4o). The Miro[1] board for each participant included the following elements: (1) A display of each Card and their corresponding index. (2) Three design tasks and the requirements. (3) A design process template was provided to offer a section for participants to document their design ideas and outcomes.

Given the diverse design backgrounds of the participants and the variety of design method cards, we provided three distinct design tasks, all requiring participants to produce design sketches and descriptions of their design outputs: (i) Smart Home Security Camera: Design a security camera for young families with children, focusing on safety and convenience. (ii) Time Management App: Design an app aimed at helping students manage their schedules. (iii) Public Bench: Design a public bench for parks or squares, intended to provide comfort and relaxation for local residents.

After participants selected one of the design tasks, they were required to select, understand, and apply design method cards to complete the design task. During the design process, participants were free to use ChatGPT (GPT-4o) for assistance and were encouraged to engage in think-aloud protocols by verbalizing their thoughts and decision-making processes. Upon completion of the study, participants received a compensation of approximately 10 USD for their participation.

To analyze the data, we first transcribed the audio recordings and extracted relevant content from each participant's Miro board to support the analysis. Afterward, we identified and categorized significant themes from the transcriptions and Miro contents. These initial themes were then discussed and refined through an iterative process, which resulted in the final and consensus-based themes.

[1]https://miro.com/

3.3 Findings

In the use of design method cards, designers typically follow a multi-phase process that involves selecting cards, understanding and applying the method to their design task (Figure 2). Through our observations, we identified several challenges designers face and required support when using the SUTD-MIT Cards with generative AI.

3.3.1 Challenges in Selecting Design Method Cards. Although SUTD-MIT Cards categorized the cards into different design phases and specified the purpose of each card, designers mentioned different levels of difficulty. Some designers (T2, P1) even spent over a quarter of their time selecting cards for their tasks (Figure 2).

During the card selection, most designers started by identifying potential methods from the design method cards based on the phase of the design task. They then excluded options that are unsuitable for the current task, considering the feasibility and relevant design domain of each method. Finally, they selected methods that align with task requirements. However, designers found *"the card method is quite general, not concise enough" (T1).* This makes it challenging for designers to determine which methods are most effective in the current situation, increasing decision-making uncertainty. P1 remarked, *"The cards don't clearly indicate when to use them, so I'm not sure how to choose the right one."* Many designers (P1, T1) felt that the cards offered limited support in decision-making and lacked detailed selection criteria.

3.3.2 Challenges in Applying Design Method Cards and Enhancing Usability with Generative AI. During the card use and understanding stage, designers pointed out that design method cards presented

standard guidance and examples for various methods, and the guidance provided often only outlines the basic framework or steps. This made it difficult for designers to obtain applicable details and insights from these conceptual guidelines. As T2 mentioned, *"Applying their design method guidance to our task and requirements is very difficult."* On the other hand, the examples on the cards did not align with the current design task. These examples cannot be mapped to the problems faced by designers, making their reference value limited and unable to provide adequate applicability. T3 and P1 also noted, *"The guidance isn't actionable"* and *"the card content lacks specificity."* Because of these issues, designers find it challenging to use these cards effectively to address real design problems. Therefore, designers use ChatGPT to generate more specific and tailored content, as T2 noted, *"We used ChatGPT to make the card's descriptions more specific."* which helps them better understand and apply the design methods. When using design method cards, designers view generative AI as a supplementary tool. They used generative AI to generate design results or expand existing ideas to further refine their design solutions. As P3 remarked *"I felt my ideas might be somewhat limited, so during mind mapping, I shared my ideas with ChatGPT to have it generate additional results."*

3.3.3 Challenges in Data Limitations and Extracting Data-Driven Insights. During the output stage, designers used the SUTD-MIT Cards to make decisions that relied on data related to designer needs, market trends, and other relevant data. However, the effectiveness of certain methods depends on the availability of specific data. Without this data, designers may struggle to apply the methods accurately. In many cases, designers generate required data using generative AI. As T1 remarked, *"I use ChatGPT to help with the lack of user data in design methods practice."* Designers found that this generated data provides practical, objective information, allowing them to make more accurate design decisions based on reference data rather than assumptions or intuition.

Some designers recognize the subjective limitations in their design thinking processes. P2 believes that personal experiences and habits often shape their design ideas and problem definitions, leading to a narrow perspective. To address these limitations, designers use ChatGPT to generate unbiased and diverse user data. This helps them create more comprehensive user profiles and overcome cognitive biases, as P2 remarked, *"My thinking when creating personas is often shaped by the people I interact with regularly, so I use generative AI to generate more diverse user data and explore different perspectives."*

3.3.4 Challenges in Design Method Card Interactivity. Designers emphasize the need for greater interactivity and adaptability in design method cards to make them more engaging and contextually relevant. P1 expressed frustration with static design method cards, mentioning that, *"I wish the cards were more interactive, rather than just being static. In reality, having a stack of cards is not much different from flipping through reference materials."* design method cards exist in static paper or digital formats and typically do not support interactive understanding or dynamic adjustments. This fixed presentation limited designers' ability to understand deeply and apply flexibly the methods.

While tools like ChatGPT provide quick responses to queries, designers face challenges with the overwhelming amount of information provided. As T2 remarked, *"Compared to traditional search, I find ChatGPT's answers more comprehensive but still difficult to navigate. I have to sift through a lot of content to find what fits my design context."* This added cognitive load highlights the need for more interactive, context-aware design method cards that can adapt to specific project context. This interactivity would not only streamline information retrieval but also reduce the burden of filtering excessive content, thus improving both the efficiency and quality of design work.

3.4 Design goals

In our formative study, we observed key challenges and support required from designers when using design method cards. Designers experienced difficulties in selecting appropriate cards due to the generic design method. Additionally, applying card guidance and example was challenging, as they were not tailored or actionable to the current design context. Designers also encountered limitations in data availability, which limited the effective use of certain methods. To address these difficulties, many designers turned to generative AI to generate more specific, actionable content and supplement necessary data, designers generally expressed a positive attitude towards the use of generative AI as a supplementary tool during the design process. Lastly, the static format of the cards limited interactivity, highlighting the need for more dynamic and context-aware cards to reduce cognitive load and improve usability. Based on these observed insights, we defined three design goals for supporting the intelligent design method cards:

DG1. Adaptable and Actionable Design Methods. Provide tailored recommendations for selecting design methods. Refine these design methods to enhance their flexibility, making them more adaptable to different design projects. Additionally, offer detailed examples to illustrate how the methods can be applied effectively, making them more relevant and actionable for designers.

DG2. Supportive Knowledge and Data. Offer comprehensive support for design method practices by integrating relevant knowledge and data. Expand the accessibility of information to enhance research depth and inform decision-making throughout the design process.

DG3. Interactive and Dynamic Card Engagement. Develop the interactive elements and dynamic capabilities of cards to promote designer engagement. Focus on delivering relevant content which reducing cognitive load.

4 Development of I-Card

4.1 Framework of I-Card

In this section, we introduce Intelligent Design Method Card Deck — I-Card, a digital design method card deck that integrates the use of generative AI. Figure 3 shows the framework of I-Card. This framework aims to support the usability, informational depth, and user experience of design method cards, thereby improving the applicability of the cards in design projects. I-Card comprises five types of cards: Info Card, Method Card, Solution Card, QA Card, and Resource Card.

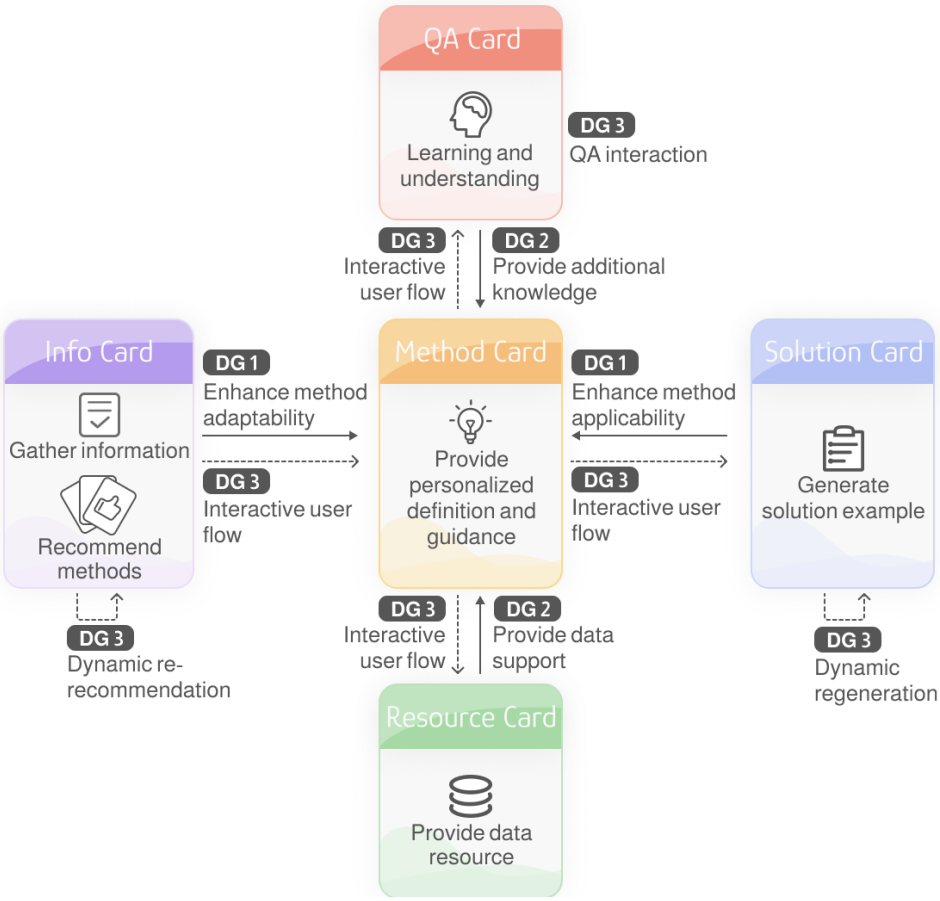


Figure 3: The framework of the proposed card deck.

To build our card framework, we began with the presentation of design methods, as the design method forms the core of the design method cards. However, presenting design methods with multiple supports in design method cards may expose itself to the risks of overcomplication, such as information overload, difficulty in navigation, and potential confusion for users [44]. To balance the challenge of providing adequate support to design methods without overwhelming the designers, we drew inspiration from classic design method cards [1, 3, 7] and created the Method Card to present only design methods. We then positioned Method Card at the center of our card framework's hierarchy. Based on support needs identified from the design goals, we derived four supporting cards that extend outward from the central Method Card, forming a cross-shaped structure as shown in Figure 3. Together, these cards form the I-Card framework. In the following paragraphs, we describe how the research team analyzed the card framework, guided by three key design goals.

To support adaptable and actionable design methods (DG1), I-Card is designed with five interconnected card types, it provides a structured yet flexible process for designers to follow. First, during the use of the cards, an input interface is required to collect personal and design-related information from designers. Info Card serves as the initial entry point of the card deck, where designers input project-related information and create a user profile. This facilitates better recommendation and generation of subsequent cards. Based on the input information from Info Card, Method Card dynamically adjusts and provide personalize design method definition and guidance, thereby enhancing Method Card adaptability by tailoring the design context. The Method Card provides detailed guidance for each step but does not show the results of method implementation. Instead, it focuses on offering personalized content to help designers better understand the processes and apply them in their design project. This approach addresses the problem of vague and unclear instructions in design method cards that many designers have pointed out. Although Method Card provides clear step-by-step guidance, designers often require more concrete examples to understand the potential outcomes of each step in practice. To address this need, we developed Solution Card. Solution Card enhances applicability by transforming the Method Card into concrete examples, it builds upon the guidance outlined in Method Card and illustrates how each step of the design methods is applied in the current design project. This gives designers a clear view of the outcomes and provides practical insights that help them translate theoretical methods into actionable design practices.

To ensure that designers have access to comprehensive and relevant information support (DG2), it is essential to provide a mechanism that can be utilized across various design methods and at different phrases of the design process. This design goal is divided into two parts: real-time knowledge and question feedback, and data support with decision-making tools. For the first part, we drew inspiration from dialogue-based tutoring systems, which have been shown to enhance learning through question-and-answer interaction [10, 50]. Based on this insight, we designed a QA (question and answer) Card to provide designers with additional knowledge, helping them obtain personalized and precise answers during the design process. It can provide adaptive answers in real-time based on the user's questions and requirements. For the second part, the data support designers need is not merely about information collection but rather about accessing data closely related to specific design method, tailored to the project's background and requirements. This allows the data can be integrated into the design process and used for data-driven decision-making. To address this, we developed a Resource Card, which provides predefined, structured, and method-specific data resources. By providing relevant data, Resource Card enables designers to explore potential design opportunities, thereby facilitating implementation of design solutions.

To create a dynamic and interactive design process (DG3), I-Card starts with an Info Card, which collects personal and design project information and leads to the selection of a Method Card. The Method Card provides personalized guidance for the design process. Next, the Solution Card offers concrete examples based on the Method Card's guidance. Supporting these main steps are the QA Card and Resource Card. QA Card gives real-time answers to specific questions, while Resource Card provides useful data for analysis and decision-making. I-Card is designed as a digital card deck integrated with generative AI, intended for use on collaboration platforms. The cards can be generated and utilized based on user backgrounds and design projects, with real-time content updates. Users can review, provide feedback, and interact with these cards on the platform, making the design process more flexible and effective. To develop and evaluate this system, we implemented a prototype in a web interface, detailed in Section 4.3.

This hierarchical card framework ensures that the individual cards avoid redundancy in functionality—each card serves a distinct role, preventing any one card from carrying too much information. Furthermore, the structure of this framework fosters more flexible and coherent interactions between the cards themselves and between the cards and the users. For instance, all five cards share data, users do not need to re-input information or rephrase their queries when transitioning from one card to another.

4.2 Design and Application of I-Card

We designed five types of cards (Figure 4) to cover various design methods, each card features a double-sided design, with the front side displaying the card's title and type and the back provideing detailed content.

4.2.1 ***Info Card*** *for Adaptive Recommendations and Selection Guidance.* The Info Card is used for inputting and managing current design information. On the back of Info Card, designers can enter their personal information (Figure 4h) and design project information (Figure 4i) into the input panel. Personal information includes the designer's experience and design field. Design project information includes the project description, design phase, goals and challenges, target users, and constraints.

Once the Info Card is filled in, designers can click the "Generate My Info Card" button, which triggers a slot-filling process. In this process, Info-Card automatically classifies the designers input into the correct fields, particularly for design project information inputs. This reduces cognitive load by allowing designers input freely and minimizing the uncertainty designers have about what to fill in each field. Based on the information inputed in Info Card, it filters and recommends three design method cards that best match the designers's requirements and provides explanations for why each method is suitable. Designers can choose the most suitable method or use the re-recommendation feature to get new suggestions based on updated information or the same input conditions.

4.2.2 ***Method Card*** *for Adaptive Method Generation.* Based on the design project information filled in the Info Card, Method Card generates personalized design method definition (Figure 4j) and guidance (Figure 4k). On the back of Method Card, it includes detailed information on each method, such as step breakdowns and applicable scenarios. Additionally, designers can interact with Method Card content by box-selecting specific steps to reveal more in-depth explanations or alternative approaches in QA Card, allowing for a more flexible and context-sensitive design process.

4.2.3 ***Solution Card*** *for Actionable Guidance.* Solution Card provides method practice results based on the Method Card (Figure 4l). On the front side, it displays the name of the innovative solution and design challenges of the design project. On the back side, it presents implementation examples that align with the design project, helping designers understand how the chosen method can be applied in real-world scenarios. Designers can click the "Regenerate" button or refine the solutions based on further inputs or adjustments in Info Card, ensuring that the practice remains relevant and adaptable throughout the design process.

4.2.4 ***QA Card*** *for Expanded Knowledge Access.* QA Card features two types of questioning: Open-Ended Questioning and Template-Based Questioning, each designed to provide relevant knowledge support and guidance based on the design context.

Template-Based Questioning: Based on patterns observed in how designers ask questions to ChatGPT during formative study, the QA Card provides pre-set question templates to streamline the querying process for designers. The question templates are divided into three categories (Table 1).

Open-Ended Questioning: This design allows designers to ask any open-ended questions by typing or using the audio-to-text feature (Figure 4n). The QA Card responds with relevant answers based on the question and the design requirements. This approach offers flexibility for designers who are exploring the design method or need quick access to information.

4.2.5 ***Resource Card*** *for Comprehensive Data Integration.* We categorized design data by analyzing the input types required by the design methods collected in Section 4.3. Specifically, we examined the key inputs these methods rely on and identified three primary

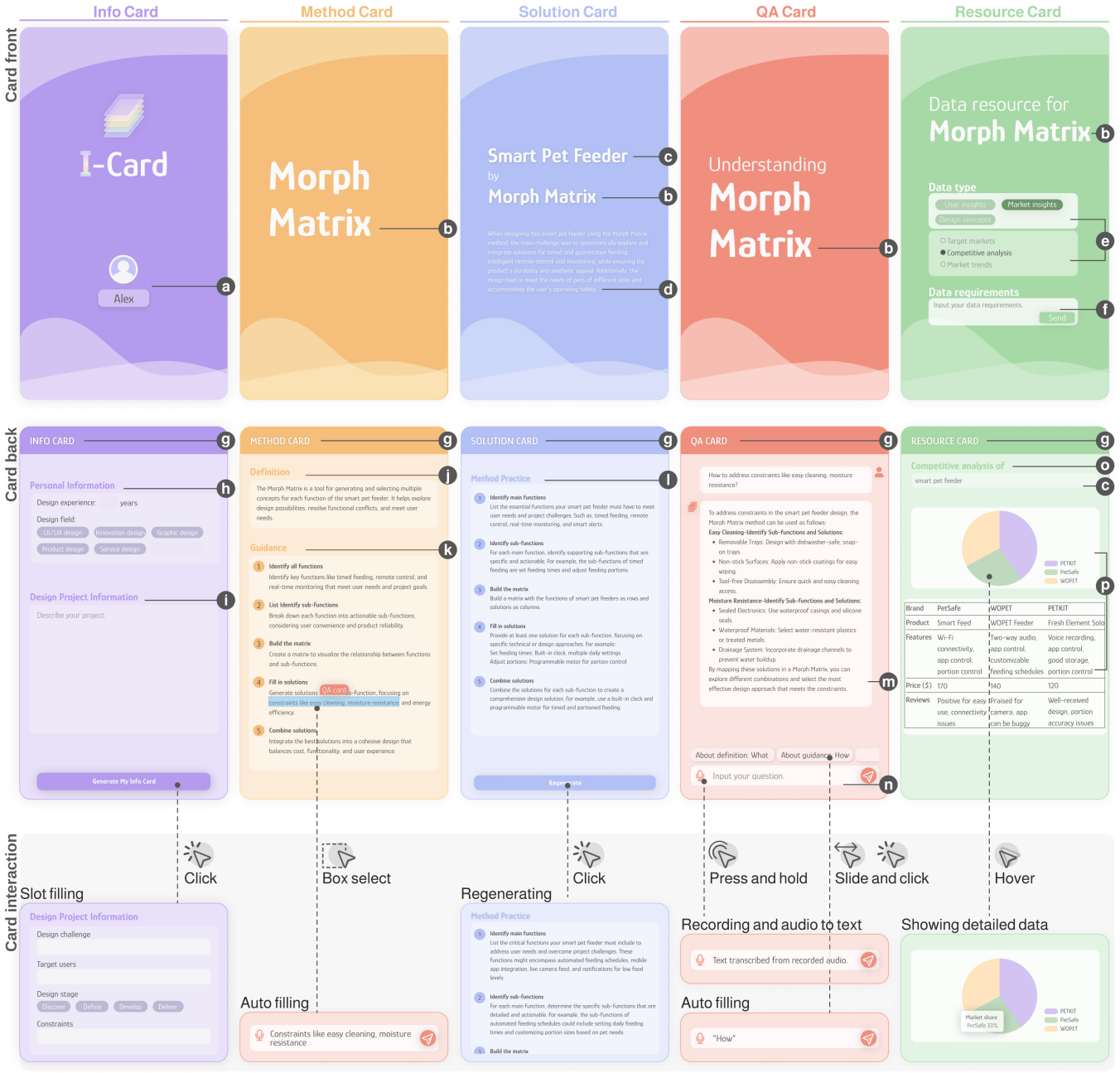


Figure 4: An intelligent morph matrix card deck. The intelligent design method card deck includes: (a) User ID; (b) Design method name; (c) Design project name; (d) Design challenge description; (e) Data type selection area; (f) Data requirement input area; (g) Card type; (h) Personal information input area; (i) Design project information input area; (j) Personalized design method definition; (k) Personalized design method guidance; (l) Design method practice results; (m) Answers to questions; (n) Question input area; (o) Chosen data type name; (p) Chosen data display.

types of data necessary to support design tasks: User Insights, Market Insights, and Design Concepts (Table 2). To further refine this categorization, we drew on resources from the four design activities outlined in Translational Resources [16].

On the front of the Resource Card, designers select the required data type from these categories (Figure 4e). They then input specific data requirements (Figure 4f), such as target demographics for User Insights or competitive landscape for Market Insights. After providing the necessary information, designers click "Send," and

Table 1: Overview of QA Card questioning templates.

Template Type	Strategy	Description
Definition	Understanding Concepts	Provides clear explanations of key concepts and extended definitions to help designers grasp core ideas and foundational principles.
Guidance of Implementation	Mastering Steps	Breaks down each step in detail, explaining the execution process and highlighting crucial points to ensure accurate application of methods.
Guidance of Reflection	Case Analysis	Provides background, outcomes, and analysis of relevant cases to help designers learn from successful or failed examples and reflect on their decisions.

Table 2: Categorization of design resources in Resource Card.

Category	Subcategory	Description
User Insights	User Behavior Data	Includes operational data, behavior patterns, and interaction records of users to analyze how users interact with the design solutions.
	User Needs	Provides user feedback, desires, and expectations to inform design decisions based on functional, emotional, and cultural requirements.
	User Psychology	Covers the psychological states, motivations, and attitudes of users to better predict user behavior and preferences.
Market Insights (Multimodal)	Market Trends	Provides data on market trends, industry dynamics, and technological advancements to help designers understand market changes.
	Competitive Analysis	Analyzes competitors' products, strategies, and market positioning to help designers identify market opportunities and threats.
	Target Market	Provides data on the size, distribution, and potential opportunities of the target market, supporting market positioning and strategy development.
Concepts and Ideas (Multimodal)	Design Concepts	Offers detailed descriptions and related research on different design concepts, helping designers explore and evaluate creative directions.
	Innovative Solutions	Showcases innovative design solutions and technologies to inspire new design ideas.
	Concept Maps and Prototypes	Provides visual concept maps, sketches, and prototypes to help designers better understand and validate design ideas.

on the back of the Resource Card, the relevant generated data is displayed, ready to be applied directly to the design project.

4.3 Technical Implementation of I-Card

This section outlines the technical implementation of I-Card, focusing on how design method is sourced, processed, and utilized across different cards to support the adaptability and interactivity of the card deck. The overall pipeline for the implementation of I-Card is illustrated in Figure 5.

4.3.1 Source of Method and Sampling. The primary goal of sourcing was to cover a wide range of significant design methods, ensuring that I-Card could draw from authoritative resources in the field of design. To achieve this, we used the Delft Design Guide [58] as a core framework, because it provides a comprehensive and widely recognized collection of design methods. Additionally, we referenced various sources, including books, publications, design tools and methods websites. During the sampling process, we sampled design methods based on three key criteria:

C1: Source Diversity—The range of source types, from books and academic publications to websites and design tools, ensuring a broad representation of different source categories.
C2: Content Quality—The clarity and thoroughness of method documentation, with methods that are well documented, clearly explained, and provide sufficient details for replication.
C3: Avoidance of Redundancy—The elimination of methods that are duplicates or very similar in content, even if they are labeled differently, ensuring a diverse set of unique methods.

4.3.2 Coding. Once the design method list was confirmed, we implemented a manual coding process to categorize the methods for recommendation and resource data classification in I-Card. The coding scheme was developed by the research team based on the Double Diamond Model [19], which provides a clear structure for

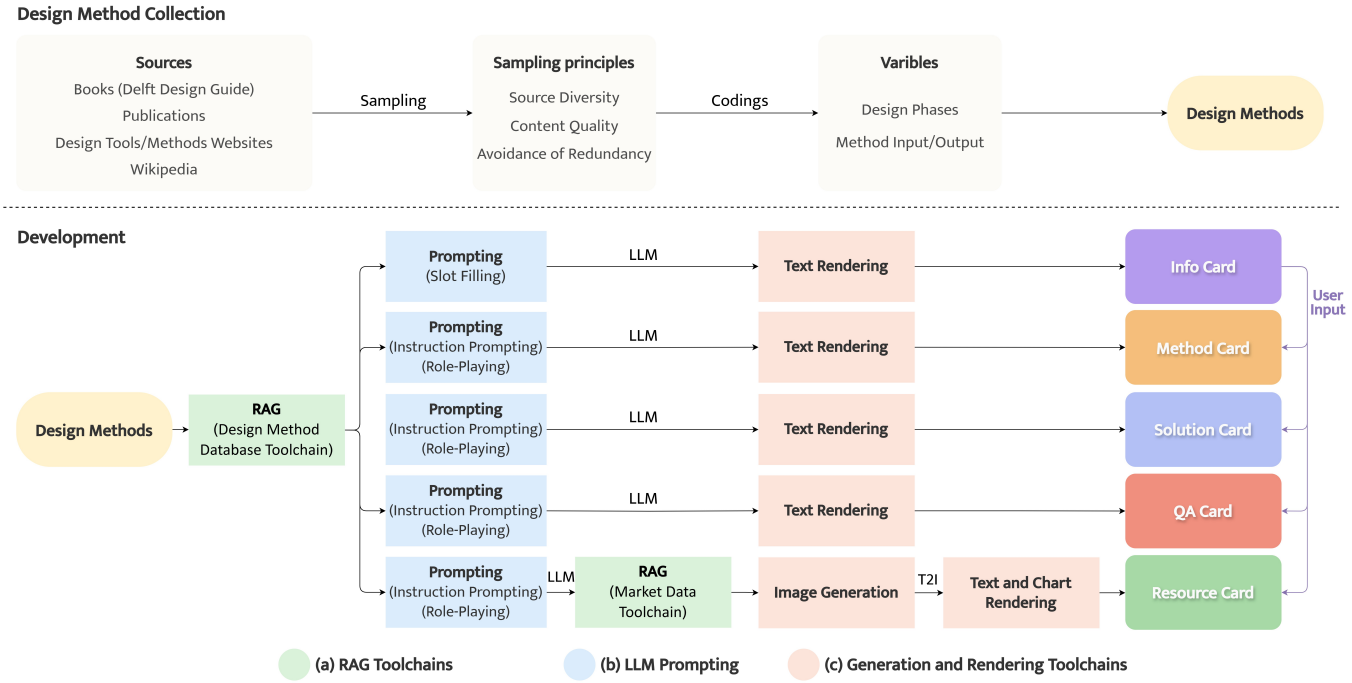


Figure 5: An overview of the technical implementation pipeline for developing I-Card. *Top:* Collecting and codeing design methods from diverse sources; *Bottom:* Applying (a)RAG toolchains and (b)prompting with LLMs to produce five cards via (c)generation and rendering toolchains.

organizing design processes. While this model guided our overall categorization, the actual classification of design methods was done manually by the research team, rather than using an existing predefined system. This approach ensures that our classification is both grounded in established design principles and suited to the specific goals of I-Card. To ensure reliability, the coding scheme was reviewed by three design methodology experts. They provided feedback on the initial classification, leading to multiple refinements. The research team iteratively adjusted the categorization based on expert input, ensuring alignment with established design principles and practical applicability in I-Card. Any inconsistencies were resolved through team discussions and consensus.

Specifically, the coding process focused on two main categories: **Design Phases** and **Method Input/Output**. Each design method from the list was reviewed and manually coded to identify its key characteristics:

- **Design Phases:** Each design method was coded into one of the four phases—Discover, Define, Develop, and Deliver of the Double Diamond Model [19], based on its main objectives and the key activities it emphasizes within the design process. This categorization helps Info Card in recommending appropriate methods based on their current phase in the design process. The Double Diamond model was chosen for its widespread acceptance and clear distinction between divergent and convergent thinking in design.
- **Method Input/Output:** To enhance the functionality of Resource Card, each method was also coded based on its input requirements and expected outputs. This coding scheme includes nine distinct types as shown in Table 2 (subcategory). For example, certain methods may require user insights (input) and produce a set of potential design concepts (output). This classification ensures that Resource Card suggests relevant data and tools tailored to the specific needs of each method.

4.3.3 Development. To build an interactive and user-friendly digital card deck, we developed a website prototype that allows switching between different design method cards, ensuring that the card deck can be widely deployed and adapted to various application scenarios. For the frontend, we chose the Vue.js[2] framework to guarantee responsive and intuitive user interactions. On the backend, we implemented Node.js[3] to manage system logic and efficiently handle multiple simultaneous requests. By integrating multiple large language models (LLMs) and services through APIs, the system dynamically generates context-aware responses and design suggestions. The design methods collected are then input into a combination of **LLM Prompting**, **Retrieval-Augmented Generation (RAG) Toolchains** and **Generation and Rendering Toolchains** to generate each card.

Two **LLM prompting** techniques are applied to generate outputs tailored for different card purposes: slot filling and instruction

[2]https://vuejs.org/
[3]https://nodejs.org/

prompting combined with role-playing (Figure 5b, prompt details in Supplementary Material).

To mitigate hallucinations often associated with LLMs, we developed two **RAG toolchains** (Figure 5a, prompt details in Supplementary Material):

- **Design Method Database Toolchain:** To ensure the scope and content of the design methods remain aligned with the list we previously collected and coded, we integrated the COZE[4] service into this toolchain. This toolchain utilizes semantic retrieval to enhance search capabilities by understanding relationships between words and sentences in a human-like manner. When the user provides input via the Input Card, the system uses the RAG method to semantically match the input with relevant design phases and the input/output variables of the methods. RAG identifies the top 10 design methods most relevant to the user's input based on semantic similarity. The LLM within RAG conducts a deeper analysis to evaluate how well each method aligns with the user's specific design context and requirements. Finally, the system recommends the three most suitable design methods based on this refined evaluation.
- **Market Data Toolchain:** For providing real-time, reliable data such as market insights, we designed and implemented a search engine-based RAG toolchain. This toolchain extracts key information from user inputs and data requirements, searches for relevant, up-to-date data via the Bing[5] search engine, and integrates the data into Resource Card.

To further enhance multimodal capabilities of I-Card, we developed three **Generation and Rendering Toolchains (Figure 5c)**:

- **Image Generation:** I-Card uses image prompts extracted by the large language model, combined with a fine-tuned DALL-E 3 model to generate images. These images are dynamically rendered on the frontend, allowing users to visualize design concepts or solutions as they interact with the cards.
- **Chart Rendering:** I-Card transforms data retrieved via the RAG toolchain into JSON arrays, which are then rendered using ECharts.js[6]. This enables complex data to be displayed visually through interactive charts, helping users better understand data trends and insights relevant to their design projects.
- **Text Rendering:** For textual content, we render the model's output in Markdown, adjusting the display format based on the specific card type to ensure the text is presented in a format that aligns with the design method card.

5 Design Feedback and Iteration

To improve the effectiveness of I-Card in achieving the proposed design goals, we gathered feedback from design students. The aim is to facilitate the rapid iteration of the design, information, and layout of I-Card.

[4]https://www.coze.com/
[5]https://www.bing.com/
[6]https://echarts.apache.org/

5.1 Method

5.1.1 Recruitment and Participants. We re-engaged five participants from the formative study because their prior experience made them well-suited to evaluate whether the issues with design method cards had been addressed and if the design goals were met. Additionally, we shared recruitment materials within the same design-focused communities used in the formative study to attract new participants, aiming to gain fresh perspectives on I-Card. Our final sample included 10 design students: five returning participants from the formative study and five new participants. The participants ranged in age from 20 to 26 years (M = 23.00, SD = 2.61) and included five males and seven females, as identified by themselves. Five participants had between three and five years of design experience, four had more than five years, and one had less than 3 years. Furthermore, one participant had previously used design method cards in a school design workshop, while three others had encountered them in academic literature, books, or social media.

5.1.2 Procedure. We conducted one-on-one, in-person design feedback sessions to gather feedback on the function, content, and layout of I-Card, as well as participants' overall perceptions of using I-Card. After each session, we revised I-Card based on the feedback received. The updated version was then tested in subsequent sessions to gather further feedback. Each session included one returning participant from the formative study and one new participant. This iterative process continued until no substantial new feedback was provided. The study procedure is illustrated in the Figure 6.

(1) On boarding (10 minutes): Participants were introduced to the research background of design method cards, our design goals, and a high-level overview of our study.

(2) I-Card Tutorial (15 minutes): Participants were tutored on the functionality, information structure, and interactive features of the five card types in I-Card. Since the input of designer and project information is necessary to generate the cards in I-Card, a virtual persona and design project were provided. The virtual persona represents a designer with specific characteristics, and the design project provides context for the task. Specifically, the virtual persona provided was an industrial product designer with five years of experience, and the design project involved creating an office chair that supports both sitting and lying down. Participants were then given the opportunity to explore I-Card independently.

(3) Quick Design Activity (40 minutes): To evaluate participants' perceptions of the practical use of I-Card, participants were asked to complete a quick design task using I-Card. Their design processes were documented on a Miro board template, and they were encouraged to engage in think-aloud protocols about their use of I-Card and the design activity. To ensure comprehensive feedback, the design tasks covered UI, product, and UX design fields, and participants could choose one task. Specifically, the UI task involved designing a plant care assistant app to provide personalized plant care advice, making the process easier for non-experts. The product design task involved designing an adjustable book stand for children that could accommodate their growing height and varying comfortable viewing angles while securely holding different book sizes. The UX task involved designing a speaker system for young urban professionals that would suit various environments

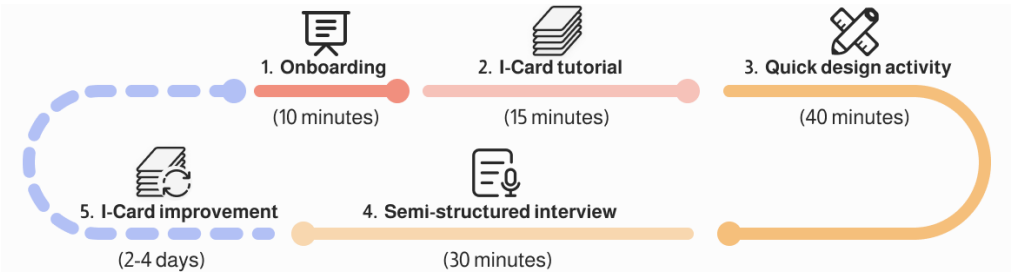


Figure 6: An illustration of the design feedback and iteration study procedure. The study consists of five stages: (1) Onboarding; (2) Tutorial; (3) Design Activity; (4) Interview; (5) I-Card Improvement.

like homes, offices, and cafés, this system needed an innovative appearance and a user-friendly interface, with the appearance being either stylized or analogical to appeal to users who value design and lifestyle quality.

(4) Semi-Structured Interview (30 minutes): We conducted a semi-structured interview where participants critically assessed each I-Card's function, content, and layout, identifying strengths, weaknesses, and areas for improvement. This feedback went beyond the think-aloud feedback collected during the design activity.

(5) I-Card Improvement (2-4 days): We made iterative improvements to I-Card design based on participants' feedback and rapidly developed new versions for the next round of iteration during the on boarding, tutorial, and design activity stages.

All stages were recorded and transcribed for thematic analysis [6] following the same approach used in the formative study. In reporting our findings, we refer to the participants as **'FP'** (**F**eedback and Iteration **P**articipants).

5.2 Findings

The main changes were on the function and content in I-Card, with some changes to layout modifications. Feedback about personal preferences, such as card color or font style, was only addressed if another participant mentioned it. Figure 7 illustrates the iterations using Solution Card as an example. The feedback and iteration process lasted for five rounds; by the fourth round, only minor layout-related feedback remained, and no further feedback was received by the fifth round. Table 3 provides an overview of the evolution of the cards over different iterations. In response to participant feedback, corresponding developments were implemented, including frontend and backend adjustments discussed in Section 4.3.3, as well as updates to the RAG toolchains and prompting. Details of these implementations are provided in the Supplementary Material. In this session, we will illustrate the changes to card function in detail, including a brief overview of the changes to card content and layout.

5.2.1 Changes to Card Function. We made several revisions to the I-Card design to improve its ability to effectively transfer design method knowledge into practice. Our goal was to enable designers to use I-Card with greater efficiency and precision. Key changes included improving the accuracy of inputs in Info Card and providing users with more control over the generation of design solution in Solution Card.

Participants noted that the slot-filling method for entering information (Figure 4i) made it easier to personalize subsequent cards. However, the lack of transparency regarding the final inputs led to confusion during usage. For instance, FP3 mentioned that after inputting personal and design project information, they wanted to review how their input was reflected in the design project information to ensure it aligned with their intentions.

"The recommended design method did not align with the needs of the design stage, possibly because I did not specify the stage during input. However, I also want to see the finalized information after slot filling to understand what the tool has actually processed."

In response, we changed the user flow of I-Card so that designers can view the final input after the slot-filling process, following the input of their personal and design project information. After using the updated version of I-Card, FP5 remarked that reviewing the final input allowed them to check for potential mismatches with their intentions. Additionally, the final input after slot-filling sparked new ideas, prompting them to enter more details that they had not considered initially.

"I saw that Info Card helped refine my design project information, such as specifying analogical design to biomimicry, which inspired me to create a speaker design based on water flow or other natural elements. This led me to add new information back into the system."

In response to this feedback, we added a feature that allows users to edit the final inputs in Info Card after completing the slot-filling process.

Participants also mentioned that while Solution Card provided valuable inspiration, it sometimes produced outputs that did not align with their intentions, similar to other generative AI tools. They requested more control over the generated solutions on Solution Card. For example, FP6 expressed dissatisfaction with the output at Step 3 and suggested adding an edit button that would automatically update the subsequent steps based on the changes, given that the steps are logically interconnected. FP6 also noted that when regenerating the entire Solution Card, they had specific requirements for the solution, such as making it more detailed and concise.

In response, we added an edit button to each step of Solution Card, with subsequent steps automatically updating after edits due to the logical sequence. Additionally, we implemented requirement input fields for all regeneration processes to better align the outputs with designer intentions. The changes are depicted in Figure

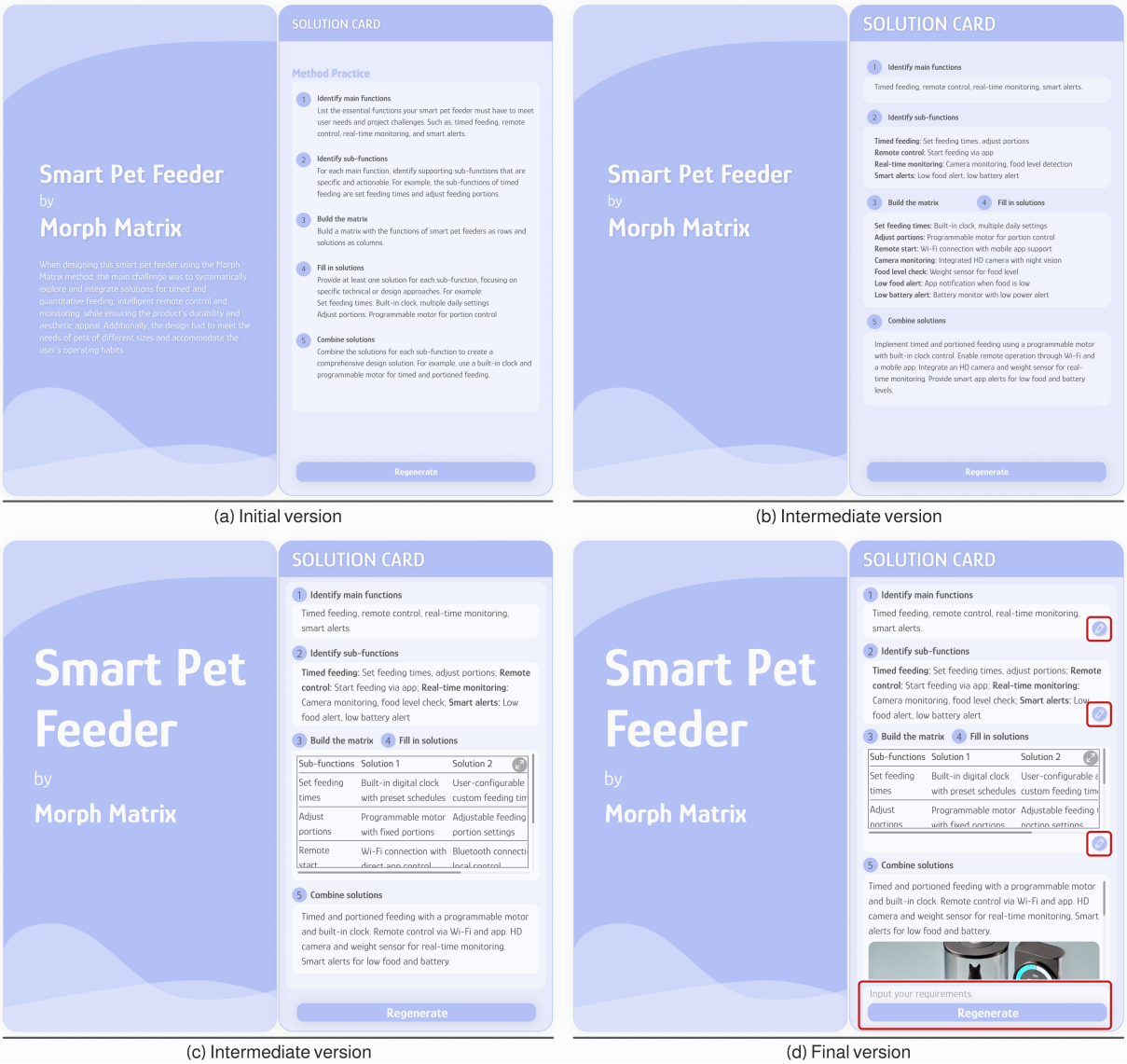


Figure 7: An overview of the iteration of Solution Card for smart pet feeder by Morph Matrix method, with modifications in (d) highlighted in a red box.

7 (c) → (d). Further details on all revisions can be found in the Supplementary Material.

5.2.2 Changes to Card Content. We revised the information on I-Card by adding theoretical information and reducing unnecessary information to enhance its authority while improving the efficiency of design method knowledge transfer. The aim was to ensure that designers could trust the personalized information and quickly derive relevant design insights. Key changes included: providing the original definition of each method on the front of Method Card, and structuring extensive data on Resource Card to reduce information overload. Full details of these changes can be found in the Supplementary Material.

5.2.3 Changes to Card Layout. We revised the layout of the cards to improve their visual usability. The aim was to utilize the visual features of the cards to present design methods in a clearer and more intuitive way. The revisions included: increasing font size, simplifying textual information, and minimizing decorative shadow effects. Full details of these changes are provided in the Supplementary Material. The final version of I-Card is shown in Figure 8.

6 User Study

6.1 Method

To explore how designers use the finalized version of the I-Card and their perceptions of its roles in improving design practice compared

Table 3: Overview of the evolution of the cards over different iterations

Iteration	Changes to card function	Changes to card content	Changes to card layout
1	· Enhanced the influence of personal information in Info Card on subsequent cards.	· Added original design method definition to Method Card for increased authority. · Differentiated the method guidance between Method and Solution Card. · Structured Resource Card information display for better readability. · Replaced the "Generate my profile" button with "Next" button label on Info Card.	· Streamlined text and added interactive elements for detailed information display.
2	· Displayed the final design project information on Info Card after slot-filling. · Added modal displays to the front of Resource Card. · Enabled requirement input for card re-recommendation.	· Aligned text and images in Resource Card. · Updated the logic of Resource Card to reduce data redundancy. · Adapted Solution Card format to information type. · Consolidated redundant steps in Solution Card.	· Increased font size for better readability.
3	· Added the final design project information editing to Info Card. · Enabled step-by-step edits in Solution Card.	· Added sources and disclaimers to Resource Card. · Included summary insights in Resource Card.	· Removed background images from Info Card. · Improved modal icon readability in Resource Card.
4	—	—	· Removed shadow effects from non-interactive UI elements.
5	—	—	—

to design method cards, we conducted a user study with sixteen designers. These designers had not participated in the formative or iteration studies, ensuring an unbiased assessment.

The participants varied in age, gender, design experience, and familiarity with design method cards. Their ages ranged from 22 to 46 years (M = 26.87, SD = 18.52). The participants included eight males and eight females, with design experience ranging from one to 24 years. Four participants had heard of design method cards, and two had previously purchased physical design method cards for use during the conceptual design phase of a design project. Participants were randomly assigned to Group 1 and Group 2.

Participants were asked to complete two design tasks using the SUTD-MIT Design Methods Cards (SUTD-MIT Cards) with ChatGPT (GPT-4o) and I-Card, respectively. The process is shown in Figure 9. First, participants were provided with an overview of design method cards, including their definition and examples, to ensure they understood this type of design support tool. Then, a tutorial was offered on the classification and information details of SUTD-MIT Cards, as well as on the information and interactions of the cards within I-Card. Participants were encouraged to explore both cards freely to become familiar with their use. After the tutorial, Group 1 used the SUTD-MIT Cards and ChatGPT to complete Task A, followed by using I-Card for Task B. Conversely, Group 2 began with I-Card for Task A and then used the SUTD-MIT Cards with ChatGPT for Task B. This arrangement was to control the effects of order and task settings on participants' perceptions of the cards. Throughout the design process, participants were instructed to think aloud. After completing the tasks, they were interviewed about the perceptions between using I-Card and the SUTD-MIT Cards with ChatGPT, its impact on their design practice, and potential areas for improvement.

The design tasks used in the user study were chosen from IDEO's open challenges, which address complex societal issues through creative problem-solving. These challenges are developed in collaboration with organizations working on real-world solutions [7]. Among IDEO's open challenges, we avoided tasks that were already developed or featured on IDEO's website to prevent participants from relying on existing solutions. From the remaining challenges, we selected two that could be addressed through a single conceptual design task, rather than broader topics requiring multiple projects or extensive time. Specifically, Task A required participants to propose design concepts for a smart food container aimed at homemakers. The design requirement was that the container must include smart monitoring functions to track food freshness in real-time and alert users when to consume it. Task B required participants to propose design concepts for a chronic skin disease care product aimed at patients needing daily at-home care. The proposed design should

[7] https://www.openideo.com/challenges

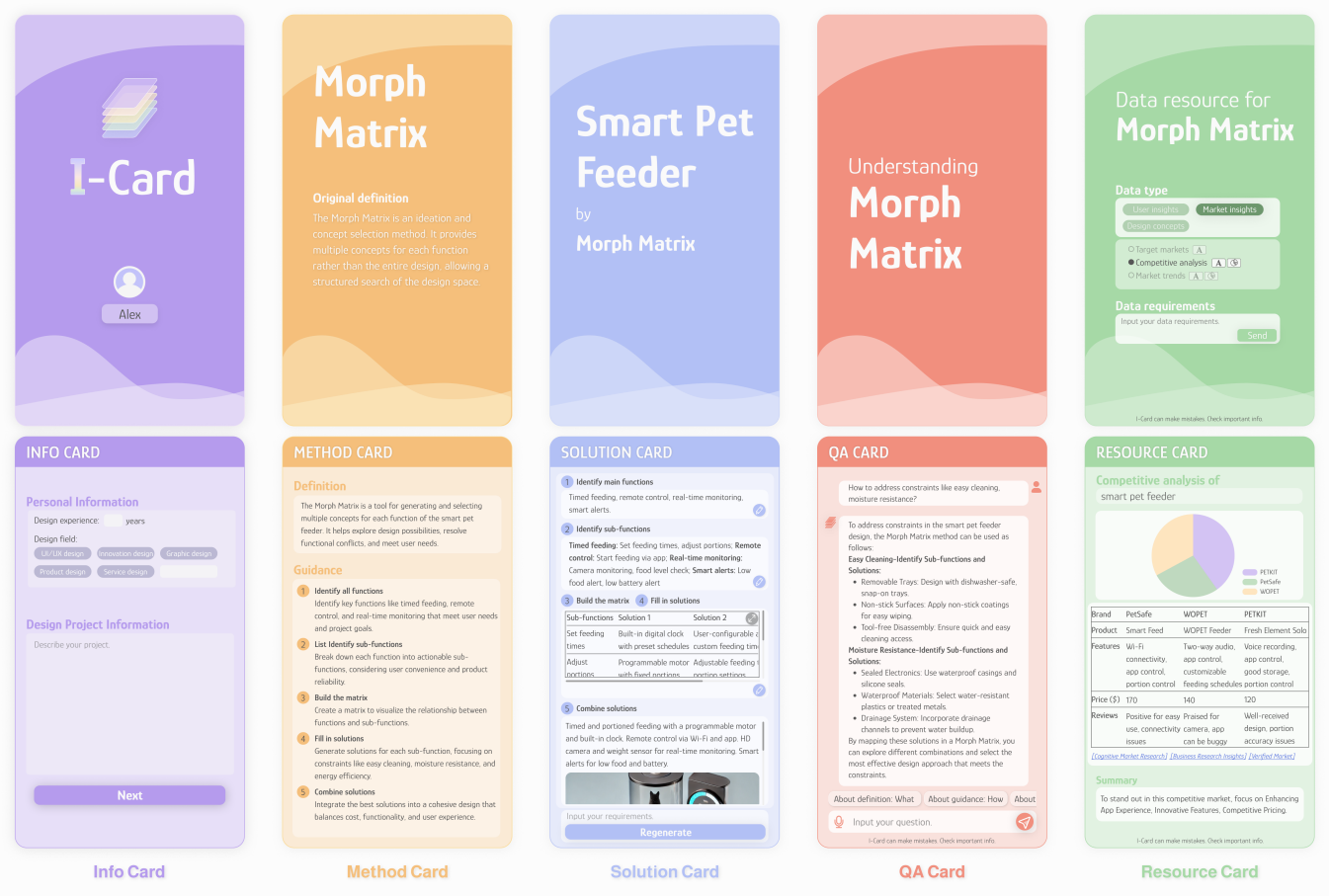


Figure 8: The final version of I-Card, including Info Card, Method Card, Solution Card, QA Card, and Resource Card.

assist in routines such as applying ointments, cold compresses, and maintaining skin cleanliness and moisture. Both tasks asked participants to produce design sketches and descriptions, with an emphasis on innovation and addressing user needs. There was no limit on the number of the produced design concepts.

6.2 Analyses

For the qualitative data, including think-aloud recordings and responses from the semi-structured interviews, we conducted a thematic analysis following the same approach used in the formative study and the design feedback and iteration sessions. In Section 6.1.3, participants are labeled as **UP**1 to **UP**8 **UP**16 (**U**ser Study **P**articipants). For the quantitative data, we recorded the design method selection time using both SUTD-MIT Cards and I-Card. The time spent inputting information into Info Card is accounted for in the design method selection time using I-Card. The entire process was fully recorded for post-study verification. Since participants were required to perform design activities with different design method cards, we applied a paired sample t-test to examine the significance of the differences in design method selection time between the SUTD-MIT Card and the I-Card.

6.3 Findings

Our findings from the user study summarize the procedures and roles of I-Card in applying design methods to design practice. We discuss how I-Card reduces barriers to applying design methods.

6.3.1 Enhancing Design Efficiency and Allowing More Focus on Design Itself. We observed that designers spent significantly less time selecting design methods using I-Card compared to SUTD-MIT Cards ($p = 0.0074 < 0.01$, $M_{SUTD-MITCards} = 5.19 \pm 2.61$ minutes, $M_{I-Card} = 2.94 \pm 0.99$ minutes, $t(15) = 3.09$, Cohen's $d = 1.18$), as shown in Figure 10. This efficiency is due to I-Card's recommendation feature, which filters three suitable design methods, allowing designers to select one without the need to manually review all available cards. UP15 stated, *"If I have to go through each card, it actually takes quite a long time to read them all. But I-Card already chose three for me."* While designers are required to input personal and design project information into Info Card, the slot-filling extraction streamlines this process, making it flexible and efficient, as UP4, UP8, and UP11 mentioned the efficiency improvements brought by the slot-filling technique. Furthermore, I-Card provides design method recommendations with reasons, enabling designers to quickly assess and decide. UP3 remarked, *"These three methods*

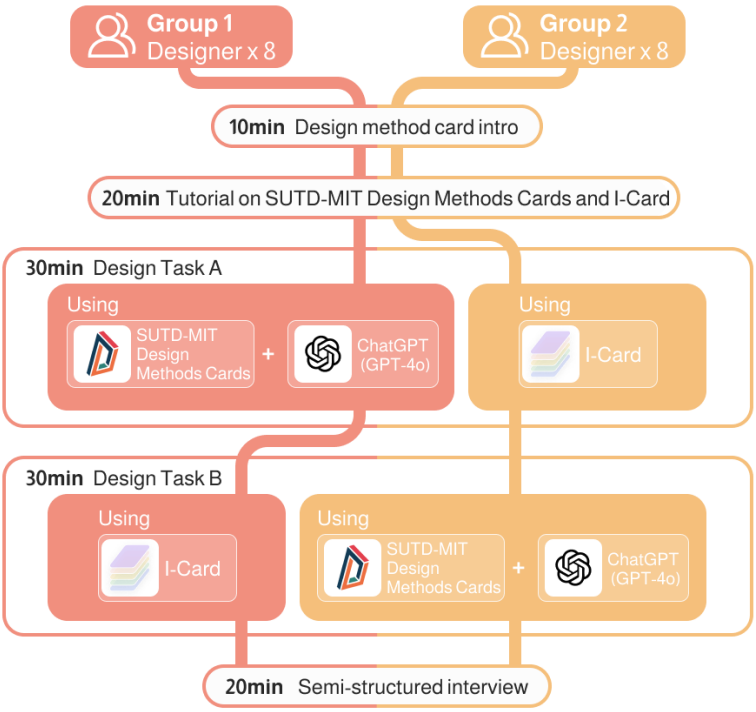


Figure 9: Overview of the procedure for User Study. In this study, we divided the sixteen participants into two groups and asked them to complete two design tasks. Group 1 initially used the SUTD-MIT Design Methods Cards and ChatGPT for Task A, followed by I-Card for Task B. Conversely, Group 2 began with I-Card for Task A and then used the SUTD-MIT Design Methods Cards and ChatGPT for Task B. After the tasks concluded, each participant took part in an one-on-one semi-structured interview.

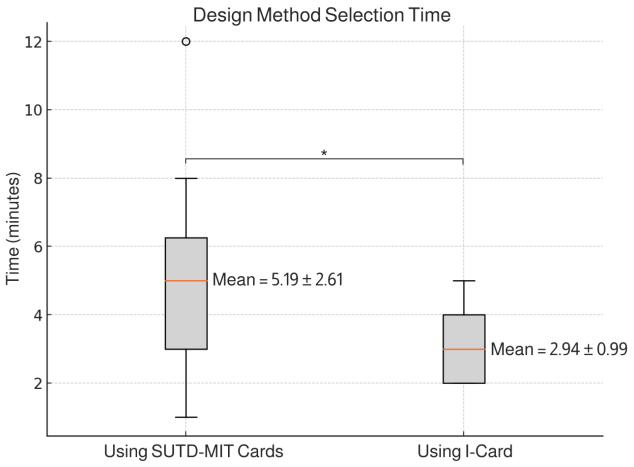


Figure 10: Comparison of design method selection time using SUTD-MIT Cards versus I-Card. * indicates significance of p < 0.01.

align well with my current situation, and each explanation clarified why they are suitable, making me want to try all of them."

Additionally, it was found that designers frequently require data, such as market or user insights, to support their design decisions or to expand their understanding of the design context. The data generation feature of Resource Card simplifies the process of data gathering. UP1, UP10, and UP12 commented, *"This green card can quickly and efficiently broaden my understanding of the entire design."* UP2, UP7, and UP11 highlighted that it *"helps me quickly acquire the necessary information."* while UP4 and UP16 added, *"It reduces the labor involved in searching and so that allows for more focus on overall system planning and decision-making."* Moreover, some participants (UP9, UP12) mentioned that the provided data type options also facilitate comprehensive and efficient data retrieval. UP9 noted, *"I might not be able to specify the exact data I need, but these (data type options) can be seen as pre-identified data needs, and I just need to look through each section."*

In addition, participants mentioned during interviews that I-Card reduced the cognitive load of using an entire card deck like SUTD-MIT Cards. First, they reported that SUTD-MIT Cards had a heavy reading burden due to their volume (UP9–UP14). Second, participants shared their reasons for perceiving I-Card as reducing cognitive load. Specifically, UP9–UP11 highlighted the benefits of the recommendation feature, while UP12 and UP13 noted that the project-specific guidance helped them construct their design process easily.

Table 4: Design method guidelines for the Morph Matrix method in both SUTD-MIT Cards and I-Card. Bold text highlights the theoretical descriptions of the design method guidelines in SUTD-MIT Cards and the personalized descriptions in I-Card.

Guidelines in SUTD-MIT Cards	Guidelines in I-Card
Identify all functions: Identify all functions of **a product, service or system from the System Model or design concepts**.	Identify all functions: Identify **key functions like timed feeding, remote control, and real-time monitoring** that meet user needs and project goals.
List functions: List functions or modules down the first column of the matrix.	List Identify sub-functions: Break down each function into actionable sub-functions, **considering user convenience and product reliability**.
Solutions to the right: Fill the columns to the right **with solutions for the function or module** in that row. Ideas can come from concept generation methods.	Build the matrix: Create a matrix to visualize the relationship between functions and sub-functions.
Map: Map the range of solutions per each function to a classification scheme.	Fill in solutions: Generate solutions for each sub-function, **focusing on constraints like easy cleaning, moisture resistance, and energy efficiency**.
Combine ideas: Combine ideas into **diverse concept variants** that seek to satisfy the specification of product, service or system.	Combine solutions: Integrate the best solutions into **a cohesive design that balances cost, functionality, and user experience**.

6.3.2 Improving the Applicability of Design Methods. Designers found that the personalized guidance in Method Card provided adaptive and actionable insights tailored to their design projects, which was especially beneficial for novice designers. Table 4 compares the theoretical descriptions of design method in SUTD-MIT Cards with the personalized descriptions in I-Card. UP2 remarked, *"For a novice like me, it's really helpful. I-Card aids in understanding the method better and applying it more effectively."* UP6 commented, *"If the method is described with vague terms, I might struggle to understand. But when it's clearly explained with my projects, it's definitely more helpful."* UP5 and UP7 also noted, *"I can generally grasp the process in SUTD-MIT Cards, but applying it still requires more learning and understanding."* However, for experienced designers, the personalized guidance mainly serves to improve efficiency. UP16 mentioned, *"I am very familiar with the whole process, so it might not be as meaningful to me."* And UP8 remarked, *"Usually, when designing, I need to prepare a sheet of paper or open an electronic page to outline the steps and processes. However, in this task, the entire process was already clearly organized for me."* It is speculated that the statics of design method cards can burden designers with the additional task of reinterpreting the method within their design project context. The static feature of SUTD-MIT Cards not only reduces efficiency but also the risk of misapplying the method. For example, when UP8 used SUTD-MIT Cards, he opted to apply parallel sketching but ended up merely replicating its form by creating superficial three-view sketches without capturing the core technique of using visual templates to explore design variations.

Additionally, Solution Card in I-Card provided step-by-step results contextualized within the design project, with each step logically building on the previous one. This gave designers a clear and specific understanding of the intended outcomes of the design methods. UP1, for example, shared, *"When using the SUTD-MIT Cards, I initially tried to use a method called TRIZ, which I had heard of as an innovative approach, but I couldn't make sense of it. The examples were too complex, so I gave up... However, with I-Card, it presented different solutions for each step. Even though I wasn't familiar with the steps of this method, I found it much easier to use."*

Moreover, the five cards in I-Card enhance both the applicability and flexibility of design methods, supporting iterative design processes. For instance, UP1 described her experience with I-Card: after receiving personalized guidance, she did not follow it strictly but selectively used what was most relevant to her process. She began with the parallel sketching method and used the QA Card to explore visual templates, which helped her decide on two potential design directions: a portable care device and a care-assist robot. These two concepts were defined early in the design process using QA Card in I-Card. She then updated her input information and applied the Product Service System Design (PSSD) method to further refine these concepts. This iterative use of different design methods within I-Card allowed her to expand and refine her ideas progressively. The design concepts proposed by UP1 using I-Card are displayed in Figure 11. This demonstrates how I-Card not only balances applicability and flexibility but also supports iterative design by enabling designers to revisit and refine their concepts, aligning with the idea that design methods should adapt to designers' evolving needs [25].

6.3.3 Providing Design Inspirations. In the user study, I-Card offered designers inspirations from various perspectives. First, Solution Card provided logically sequenced inspirations based on the design method guidance. The ability to edit each step's results and have subsequent steps automatically update improved the coherence and control over the generated outcomes. As UP3 noted, *"The step-by-step editing feature of Solution Card makes it align the inspirations generated by the LLM with the logical flow of using design methods."* Secondly, QA Card, with its flexible and responsive answers, could inspire design ideas tailored to specific designer needs. UP2 observed, *"QA Card analyzes what I'm designing and*

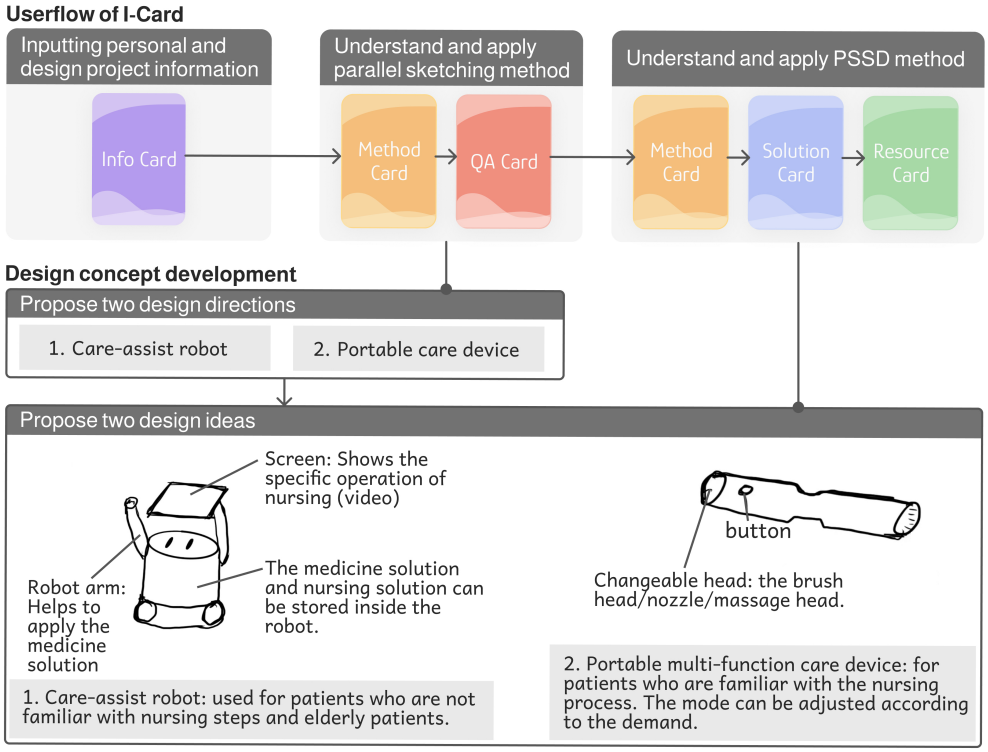


Figure 11: Userflow of I-Card and design concept development by UP1.

offers relevant design suggestions." It was observed that QA Card also complemented the framework of the generated outcomes in Solution Card. UP5 added, *"It can build more ideas on the basic framework provided by Solution Card. It expands several main ideas, which you can then enrich with detailed responses."* Finally, the design concept data within Resource Card provided concrete and organized inspiration focused on final design outputs. UP1, UP3, and UP12 commented, *"The data generated was quite inspiring; it helped me solidify ideas that weren't fully formed, and suddenly I realized, this approach could work."*

6.3.4 Supporting Design Decision-Making. The study found that the knowledge and data support provided by I-Card effectively aids design decision-making. QA Card improves designers' understanding of both design method knowledge and domain-specific knowledge, thereby enhancing their decision-making abilities when addressing challenges in those domains. UP1, UP2, UP13 and UP14 noted, *"It not only clarifies specific questions but also integrates guidance with the design method I'm currently using."* highlighting the role of QA Card in providing design method knowledge. UP8 added, *"The QA feature is particularly helpful, especially when working with unfamiliar products like care devices. It allows me to quickly gain insights into the field by searching for relevant knowledge."* emphasizing the efficiency in helping designers navigate unfamiliar design domains. Additionally, the user and market insights generated by Resource Card serve as reference data that can either support or deny current design decisions. UP6 explained, *"For example, it provides market data percentages, so when someone asks about the rationale behind my design, or when I'm considering its practicality, I have solid data to back it up."* Similarly, UP5 used Resource Card to generate user insight data showing that care products are typically used around three times a day. This insight led her to abandon the idea of a stationary care chair, thus correcting a misaligned design direction.

7 Discussion

7.1 The Strengths of Intelligent Design Method Cards

Different mediums of design method cards offer unique advantages. The tangibility of physical cards enables hands-on and seamless interaction, which stimulates creativity and encourages collaboration [40]. In contrast, digital cards offer greater flexibility, allowing for new interactions like copying, annotating, and zooming, and they are more accessible for remote or hybrid design teams due to their online availability [57]. Specifically, intelligent digital cards, like I-Card, bring additional features compared to static digital cards. In this section, we summarize the unique features of intelligent cards and discuss their strengths, including improved adaptivity, dynamics, interactivity, and additional support.

(1) **More Effective Knowledge Transfer:** With the help of generative AI, I-Card adapts design methods to both the designer's needs and the design project context by personalizing information based on designer inputs. Participants

(U1, U2, U6) are able to quickly grasp the methods when they were contextualized in I-Card. They showed a noticeable reduction in hesitancy when applying methods, as they spent less time analyzing method instructions and more time on actual design tasks (Section 6.3.1). UP5 highlighted the structure's benefits, *"Method Card and Solution Card provide different levels of guidance within the same design method. Method Card gives an overview, while Solution Card breaks it down step-by-step, which feels like a natural progression and lightens my processing load."* This behavioral shift suggests that the personalized design method also reduced cognitive overload and improved the accessibility of design knowledge which lead to a effective knowledge transfer.

(2) **Broader Applicability in Design Fields:** Unlike static cards, intelligent cards dynamically adjust to the changing challenges of design tasks, ensuring that the design methods and insights they provide remain relevant. This dynamics broaden the applicability of design methods. For example, some designers (UP1, UP6, UP8, UP2) incorporated elements from various design fields, including robotics design, software design, kitchen design, and wheelchair design. By updating project details in Info Card and exploring alternative methods (like UP1), designers were able to tailor I-Card's guidance to the specific needs of each field, and also access relevant design data and knowledge from different fields.

(3) **Enhanced Engagement and Iterative Design:** I-Card enables designers to engage in a flexible, non-linear design process. The video recordings from the user study showed that some participants iterated on their design concepts by updating input information in Info Card and applying additional design methods after reaching initial design concepts (Section 6.3.2). This iterative behavior is supported by specific features of I-Card: Solution Card, QA Card, and Resource Card provide iterative inspirations through generated solutions, answers, and data, respectively, while Info Card allows users to specify their design phase, enabling access to design methods tailored to multiple phases of the design process. Besides, its support for multimodal interactions—such as incorporating images and audio—makes the design experience more intuitive. These interactive features create a more engaging and iterative design process that aligns with the exploratory nature of design.

(4) **Efficient and Comprehensive Decision-Making:** As a generative AI-supported tool, I-Card integrates additional knowledge and data support directly into the card interface, enabling designers to make more informed design decisions with greater efficiency. This holistic support enhances the depth and rigor of design solutions, ensuring that the decisions are based on comprehensive context and empirical data.

7.2 Implications for the HCI Community

Knowledge transfer in the context of design method knowledge refers to the process of enabling designers to understand design methods and apply them to design activities, guiding the development of design solutions. This study offers a multi-layered perspective on this process, focusing on three key, sequential aspects: the design method knowledge itself, the transfer process, and design activities. The three proposed design goals are derived from these aspects. Enhancing adaptivity (DG1) improves the comprehensibility of the knowledge itself by providing personalized knowledge descriptions based on designers' input. This is achieved in I-Card through Info Card, which gathers information, as well as Method and Solution Card, which provide tailored design methods and solution examples. Enhancing interactive and dynamic engagement (DG2) focuses on making the transfer process more engaging through the interactive QA Card, dynamic editing in Solution Card, and the flexible user flow of I-Card. Providing additional knowledge and data support (DG3) strengthens the application of design method knowledge in design activities. In I-Card, Resource Card generates relevant data, while QA Card provides design method and domain knowledge. This intelligent design method card deck framework supports both the understanding and application of design method knowledge by enhancing not only the knowledge descriptions and transfer process but also its practical application. The findings contribute to the broader discussion of knowledge transfer in HCI field, extending beyond design method knowledge and card-based systems. Future research could further explore improving the adaptability of knowledge, the engagement of transfer processes, and ensuring practical support for applying this knowledge.

7.3 Opportunities to Further Support Design Method Cards

While I-Card is designed to support design method cards, its underlying framework has the potential to be expanded to other types of design method cards, such as Tooling, Human Insights, Domains, and Materials [27]. These cards could benefit from the adaptable capabilities offered by the integration of generative AI. One possible application is Intent Cards [38], which focus on influencing human behavior through design, could be placed at the center of the I-Card framework. For instance, an Intent Card like "Implied Sequences", which helps designers create interfaces that guide users through a logical flow. Info Card would gather information about the user demographics, interaction goals, and then recommend appropriate Intent Cards. Solution Card could generate concrete examples showing how designers have effectively guided users through similar design sequences. QA Card and Resource Card would answer questions and provide supporting data, such as user studies on navigation patterns, and best practices for interface layout, enabling designers to make data-driven decisions. By using the flexibility of I-Card framework, we have the opportunity to better support a variety of design activities and challenges.

7.4 Limitations

Generative AI presents challenges in the form of hallucinations. For I-Card, it affects the accuracy of personalized design methods and the reliability of data support. To address the first issue, we have incorporated design method knowledge from authoritative books as knowledge sources. For the second, we provided data source links within Resource Card, allowing designers to fact-check or explore

further details. Additionally, we included hallucination warnings on the cards. However, the impact of hallucinations remains unavoidable. Future research could focus on minimizing these effects through methods like fine-tuning the model with domain-specific data or applying reinforcement learning with human feedback to enhance accuracy.

In our formative and iteration studies, the small sample size and limited differentiation in experimental settings resulted in no significant differences between how teams and individual participants used the cards. This limitation was primarily due to the diverse nature of team collaboration models, which we could not exhaustively explore within the scope of our study. Additionally, time and resource constraints prevented us from designing experiments with significant differentiation between team and individual use. Despite these limitations, we believe they did not affect the validity of our findings. The flexibility of the final card design ensures it remains applicable to both individual and team contexts. Besides, the formative and iteration studies lacked the participation of professional designers. Although professional designers were included in the user study and reported a preference for I-Card, future work will systematically consider the differences between professional designers and novices.

8 Conclusion

I-Card, an intelligent design method card deck that aims to support the applicability of design methods has been developed and tested. This card deck offers design knowledge and data support, and improves interactive and dynamic engagement. By leveraging generative AI to generate personalized design methods based on designer input, I-Card increases both the adaptability and applicability of design methods. Additionally, generative AI is used to provide support by answering knowledge questions and generating data. I-Card's interactive features, such as QA interactions, regeneration based on inputs, as well as slot-filling technique, further enhance user engagement. A user study with sixteen designers highlighted I-Card's roles in design practice: improving design efficiency, increasing the applicability of design methods, offering diverse design inspirations, and supporting decision-making. Finally, we discuss the extensibility of the I-Card framework to other knowledge transfer tools in the HCI community.

Acknowledgments

This work was supported by the National Key R&D Program of China under Grant No. 2022YFB3303304.

References

[1] 18F. 2023. 18F Method Cards. https://guides.18f.gov/methods/
[2] Tessa Aarts, Linas K Gabrielaitis, Lianne C De Jong, Renee Noortman, Emma M Van Zoelen, Sophia Kotea, Silvia Cazacu, Lesley L Lock, and Panos Markopoulos. 2020. Design card sets: Systematic literature survey and card sorting study. In *Proceedings of the 2020 ACM designing interactive systems conference*. 419–428.
[3] Design Sprint Academy. 2024. Design Sprint Method Cards. https://www.designsprint.academy/product/design-sprint-method-cards
[4] Genrich Altshuller. 1984. *Creativity as an Exact Science: The Theory of the Solution of Inventive Problems*. Gordon and Breach Science Publishers.
[5] Clayton Berger et al. 1993. Kano's methods for understanding customer-defined quality. *Center for Quality Management Journal* 2, 4 (1993), 3–36.
[6] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
[7] SUTD-MIT International Design Centre. 2013. SUTD-MIT Design Methods Cards. https://idc.sutd.edu.sg/wpcontent/uploads/sites/10/2017/10/DesignMethod-Cards_SUTDMITIDC-2017.pdf Accessed: 2024-09-05.
[8] Liuqing Chen, Zebin Cai, Zhaojun Jiang, Jianxi Luo, Lingyun Sun, Peter Childs, and Haoyu Zuo. 2024. AskNatureNet: A divergent thinking tool based on bio-inspired design knowledge. *Advanced Engineering Informatics* 62 (2024), 102593.
[9] Liuqing Chen, Zebin Cai, Zhaojun Jiang, Lingyun Sun, Peter Childs, and Haoyu Zuo. 2024. A knowledge graph-based bio-inspired design approach for knowledge retrieval and reasoning. *Journal of Engineering Design* (2024), 1–31.
[10] Liuqing Chen, Zhaojun Jiang, Duowei Xia, Zebin Cai, Lingyun Sun, Peter Childs, and Haoyu Zuo. 2024. BIDTrainer: An LLMs-driven Education Tool for Enhancing the Understanding and Reasoning in Bio-inspired Design. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–20.
[11] Liuqing Chen, Qianzhi Jing, Yixin Tsang, Qianyi Wang, Lingyun Sun, and Jianxi Luo. 2024. DesignFusion: Integrating Generative Models for Conceptual Design Enrichment. *Journal of Mechanical Design* 146, 11 (2024).
[12] Liuqing Chen, Pan Wang, Hao Dong, Feng Shi, Ji Han, Yike Guo, Peter RN Childs, Jun Xiao, and Chao Wu. 2019. An artificial intelligence based data-driven approach for design ideation. *Journal of Visual Communication and Image Representation* 61 (2019), 10–22.
[13] Liuqing Chen, Haoyu Zuo, Zebin Cai, Yuan Yin, Yuan Zhang, Lingyun Sun, Peter Childs, and Boheng Wang. 2024. Towards Controllable Generative Design: A Conceptual Design Generation Approach Leveraging the FBS Ontology and Large Language Models. *Journal of Mechanical Design* (2024), 1–40.
[14] Peter Childs, Ji Han, Liuqing Chen, Pingfei Jiang, Pan Wang, Dongmyung Park, Yuan Yin, Elena Dieckmann, and Ignacio Vilanova. 2022. The creativity diamond—A framework to aid creativity. *Journal of Intelligence* 10, 4 (2022), 73.
[15] DaEun Choi, Sumin Hong, Jeongeon Park, John Joon Young Chung, and Juho Kim. 2024. CreativeConnect: Supporting Reference Recombination for Graphic Design Ideation with Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–25.
[16] Lucas Colusso, Cynthia L Bennett, Gary Hsieh, and Sean A Munson. 2017. Translational resources: Reducing the gap between academic research and HCI practice. In *Proceedings of the 2017 conference on designing interactive systems*. 957–968.
[17] Jaap Daalhuizen and Philip Cash. 2021. Method content theory: Towards a new understanding of methods in design. *Design Studies* 75 (2021), 101018.
[18] Richard Lee Davis, Thiemo Wambsganss, Wei Jiang, Kevin Gonyop Kim, Tanja Käser, and Pierre Dillenbourg. 2024. Fashioning Creative Expertise with Generative AI: Graphical Interfaces for Design Space Exploration Better Support Ideation Than Text Prompts. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–26.
[19] Design Council. 2024. Framework for Innovation: Design Council's Double Diamond. https://www.designcouncil.org.uk/news-opinion/design-process-what-double-diamond. Accessed: September 5, 2024.
[20] Kees Dorst. 2011. The core of 'design thinking'and its application. *Design studies* 32, 6 (2011), 521–532.
[21] Salma Elsayed-Ali, Sara E Berger, Vagner Figueredo De Santana, and Juana Catalina Becerra Sandoval. 2023. Responsible & Inclusive Cards: An online card tool to promote critical reflection in technology industry work practices. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–14.
[22] Frederic Gmeiner, Jamie Lynn Conlin, Eric Handa Tang, Nikolas Martelaro, and Kenneth Holstein. 2024. An Evidence-based Workflow for Studying and Designing Learning Supports for Human-AI Co-creation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–15.
[23] Sydney A Gregory. 2013. *The design method*. Springer.
[24] Kim Halskov and Peter Dalsgaard. 2006. Inspiration Card Workshops. *Proceedings of the DIS'06 Conference* (2006), 2–11. doi:10.1145/1142405.1142409
[25] Steve Harrison and Deborah Tatar. 2011. On methods. *Interactions* 18, 2 (2011), 10–11.
[26] Karen Holtzblatt and Hugh Beyer. 2004. *Rapid contextual design: A how-to guide to key techniques for user-centered design*. Elsevier.
[27] Gary Hsieh, Brett A Halperin, Evan Schmitz, Yen Nee Chew, and Yuan-Chi Tseng. 2023. What is in the cards: Exploring uses, patterns, and trends in design cards. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–18.
[28] IDEO. 2019. IDEO Design Cards. https://www.ideo.com/post/method-cards. Accessed: September 5, 2024.
[29] IDEO. 2024. Design Thinking for Educators Toolkit. https://designthinking.ideo.com/resources/design-thinking-for-educators. Accessed: September 5, 2024.
[30] LUMA Institute. 2012. *Innovating for People: Handbook of Human-Centered Design Methods*. LUMA Institute.
[31] Shuo Jiang and Jianxi Luo. 2024. AutoTRIZ: Artificial Ideation with TRIZ and Large Language Models. *arXiv preprint arXiv:2403.13002* (2024).
[32] John Chris Jones. 1992. *Design methods*. John Wiley & Sons.
[33] Ron Kohavi et al. 2007. Practical guide to controlled experiments on the web: listen to your customers not to the hippo. *Proceedings of the 13th ACM SIGKDD international conference on Knowledge discovery and data mining* (2007), 959–967.

[34] Chrysanthi Konstanti, Evangelos Karapanos, and Panos Markopoulos. 2022. The Behavior Change Design Cards: A Design Support Tool for Theoretically-Grounded Design of Behavior Change Technologies. *International Journal of Human–Computer Interaction* 38, 13 (2022), 1238–1254.

[35] Huisung Kwon, Yunjae Josephine Choi, Sunok Lee, and Sangsu Lee. 2024. Unveiling the Inherent Needs: GPT Builder as Participatory Design Tool for Exploring Needs and Expectation of AI with Middle-Aged Users. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–6.

[36] Christine P Lee, Min Kyung Lee, and Bilge Mutlu. 2024. The AI-DEC: A Card-based Design Method for User-centered AI Explanations. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 1010–1028.

[37] Seung Won Lee, Tae Hee Jo, Semin Jin, Jiin Choi, Kyungwon Yun, Sergio Bromberg, Seonghoon Ban, and Kyung Hoon Hyun. 2024. The Impact of Sketch-guided vs. Prompt-guided 3D Generative AIs on the Design Exploration Process. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18.

[38] Dan Lockton. 2010. *Design with intent: 101 patterns for influencing behaviour through design*. Dan Lockton.

[39] Andrés Lucero and Juha Arrasvuori. 2010. PLEX Cards: A Source of Inspiration When Designing for Playfulness. In *Proceedings of the Fun and Games 2010 Conference*. ACM Press, 28–37. doi:10.1145/1823818.1823821

[40] Andrés Lucero, Peter Dalsgaard, Kim Halskov, and Jacob Buur. 2016. Designing with cards. *Collaboration in creative design: Methods and tools* (2016), 75–95.

[41] Miranda Luong, Daragh Byrne, and Marti Louw. 2021. Makercards: Designing an electronic component discovery tool to support remote physical computing education. In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference*. 476–482.

[42] William Moore and Edgar A. Pessemier. 1987. The Concept Testing Process: A guide for developing the best possible ideas for your products. *Journal of Marketing* 3 (1987), 24–36.

[43] Jakob Nielsen. 1994. *Usability Engineering*. Morgan Kaufmann.

[44] Novia Nurain, Chia-Fang Chung, Clara Caldeira, and Kay Connelly. 2024. Designing a Card-Based Design Tool to Bridge Academic Research & Design Practice For Societal Resilience. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–20.

[45] HyunJoo Oh, Sherry Hsi, Noah Posner, Colin Dixon, Tymirra Smith, and Tingyu Cheng. 2023. Making computing visible & tangible: A paper-based computing toolkit for codesigning inclusive computing education activities. *International Journal of Child-Computer Interaction* 38 (2023), 100602.

[46] Fred Paas, Alexander Renkl, and John Sweller. 2003. Cognitive load theory and instructional design: Recent developments. *Educational psychologist* 38, 1 (2003), 1–4.

[47] John Pruitt and Tamara Adlin. 2006. *The Persona Lifecycle: Keeping People in Mind Throughout Product Design*. Morgan Kaufmann.

[48] Erika Root, Wilko Heuten, and Susanne Boll. 2019. Maker Cards: Evaluating design cards for teaching physical computing to middle-school girls. In *Proceedings of Mensch Und Computer 2019*. 493–497.

[49] Robin Roy and James P Warren. 2019. Card-based design tools: A review and analysis of 155 card decks for designers and designing. *Design Studies* 63 (2019), 125–154.

[50] Sherry Ruan, Liwei Jiang, Justin Xu, Bryce Joe-Kun Tham, Zhengneng Qiu, Yeshuang Zhu, Elizabeth L Murnane, Emma Brunskill, and James A Landay. 2019. Quizbot: A dialogue-based adaptive learning system for factual knowledge. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–13.

[51] K.S. Sanhueza and C. Nikulin. 2019. *A framework for design methods, models and techniques for product and process development: A novel classification to increase their applicability in practices*. Vol. 25. Emerald Publishing, 1759–1782. doi:10.1108/BPMJ-12-2018-0380

[52] Hasti Seifi, Sean Chew, Antony James Nascè, William Edward Lowther, William Frier, and Kasper Hornbæk. 2023. Feellustrator: A design tool for ultrasound mid-air haptics. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.

[53] Orit Shaer, Angelora Cooper, Osnat Mokryn, Andrew L Kun, and Hagit Ben Shoshan. 2024. AI-Augmented Brainwriting: Investigating the use of LLMs in group ideation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.

[54] Donghoon Shin, Lucy Lu Wang, and Gary Hsieh. 2024. From Paper to Card: Transforming Design Implications with Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–15.

[55] Erik Stolterman and James Pierce. 2012. Design tools in practice: studying the designer-tool relationship in interaction design. In *Proceedings of the designing interactive systems conference*. 25–28.

[56] John Sweller. 1994. Cognitive load theory, learning difficulty, and instructional design. *Learning and instruction* 4, 4 (1994), 295–312.

[57] Lena Tørresdal, Francesco Gianni, and Monica Divitini. 2021. The digital transformation of card-based design toolkits: From tiles to digitiles. In *CHI Greece 2021: 1st International Conference of the ACM Greek SIGCHI Chapter*. 1–5.

[58] Annemiek Van Boeijen, Jaap Daalhuizen, Roos Van Der Schoor, and Jelle Zijlstra. 2014. *Delft design guide: Design strategies and methods*.

[59] Ken Wallace. 2011. Transferring design methods into practice. In *The future of design methodology*. Springer, 239–248.

[60] Zezhong Wang, Shan Hao, and Sheelagh Carpendale. 2024. Card-Based Approach to Engage Exploring Ethics in AI for Data Visualization. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.

[61] Justin D Weisz, Jessica He, Michael Muller, Gabriela Hoefer, Rachel Miles, and Werner Geyer. 2024. Design Principles for Generative AI Applications. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–22.

[62] Christiane Wölfel and Timothy Merritt. 2013. Method card design dimensions: A survey of card-based design tools. In *Human-Computer Interaction–INTERACT 2013: 14th IFIP TC 13 International Conference, Cape Towrn, South Africa, September 2-6, 2013, Proceedings, Part I 14*. Springer, 479–486.

[63] Lixiang Yan, Vanessa Echeverria, Gloria Milena Fernandez-Nieto, Yueqiao Jin, Zachari Swiecki, Linxuan Zhao, Dragan Gašević, and Roberto Martinez-Maldonado. 2024. Human-AI Collaboration in Thematic Analysis using ChatGPT: A User Study and Design Recommendations. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.

[64] Pippa Yeoman and Lucila Carvalho. 2019. Moving between material and conceptual structure: Developing a card-based method to support design for learning. *Design Studies* 64 (2019), 64–89.